Strengthening Community Resilience in Fragile and Conflict-Affected Societies

Evidence from a Randomized Controlled Trial of a CDD Program with a Conflict Resolution Dimension in Eastern Congo

Endline Impact Evaluation Report

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The Democratic Republic of Congo

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Executive summary

This document presents a randomized impact evaluation of a Community-Driven Development (CDD) program in Democratic Republic of Congo (DRC). The CDD program – with an envelope size of $30 million – is part of the Eastern DRC Recovery Project (STEP in its French acronym), funded through a grant from the World Bank and implemented by the Social Fund of the DRC (FSRDC). The aim of the CDD program is to increase community resilience by improving access to infrastructure and by strengthening social cohesion. To verify whether this aim was met, the World Bank's Development Impact Evaluation (DIME) Department conducted a randomized control trial (RCT), together with FSRDC.

The program targeted the provinces of South Kivu, North Kivu, Bas Uele, Haut Uele, Ituri, and Tschopo. From 2016 to 2020, the CDD program was randomly assigned to about two thirds of 400 communities with eligible project proposals. The selected communities received a budget of up to $100,000 for the construction or rehabilitation of infrastructure. In addition, each recipient community of a CDD project received training to manage the project in an inclusive and participatory way. Training was given by local NGOs to existing Local Development Committees (LDCs), who were then responsible for project management and for setting up a structure for infrastructure maintenance. To promote inclusiveness, the LDC was encouraged to include a 30 percent representation of the most vulnerable (women, internally displaced and youth at risk). To promote community ownership, the community was expected to co-finance 10 percent of the project (in cash or kind).

A random half of selected CDD communities also received a conflict mitigation component, which consisted of conflict prevention and resolution activities. These activities were identified by local NGOs specialized in the matter and guided by conflict mappings that took place in all administrative territories concerned. To make the activities sustainable and sufficiently rooted in local realities, the NGOs trained existing local authorities (formal and de facto) to take on the monitoring and management of conflicts. Specific activities, in collaboration with these structures, could include among others establishing or strengthening of inter- and intra-community dialogue; and improving integration conditions for returnees, or refugees, for instance by identifying needs in terms of social cohesion and housing.

According to the theory of change guiding the intervention, the CDD program would lead not only to improvements in community infrastructure but also to more social cohesion, because of the adopted inclusive and participatory process, and the demonstration effect this entails (when it leads to a successful project implementation). The conflict mitigation component would enhance the CDD effect, by reducing internal divisions that could work against its effective implementation. In addition, the add-on component also has the potential to directly improve social cohesion by mitigating conflicts. Our impact evaluation puts this theory of change to a direct test. **Particularly, this research seeks to identify causal impacts of the CDD intervention in its basic form as well as when augmented with the afore-described conflict mitigation component** (henceforth referred to as 'CDD+'). The main evaluation questions guiding this study are threefold: ***(1)*** *Does the CDD program improve access to community infrastructure?* ***(2)*** *Does it enhance social cohesion?* ***(3)*** *Are CDD+ projects more effective in improving access to infrastructure and social cohesion?*

The outcomes of interest are situated along two main dimensions: infrastructure and social cohesion. Regarding infrastructure, we focus on the presence of infrastructure, in how far individuals have access to it, make effectively use of it, and whether it has led to improvements in health and education. To capture social cohesion, we measure (intra- and inter-ethnic) trust, participation in community organizations, perceived cleavages, the integration of outsiders (e.g., IDPs), information transmission from the chief to the community members, the nature and frequency of conflict (within and between villages), and whether and how conflicts are being resolved. Additionally, to capture the effect of the intervention on the socioeconomic wellbeing of the participants, we measured a set of **secondary outcomes**, including measures of economic welfare, income generating activities, and subjective well-being.

Two data collection rounds took place: an ex-ante round shortly before the start of the CDD project and an ex-post round, on average eleven months after the project finished. Due to insecurity, inaccessibility, and other operational challenges, 51 out of the 416 initially randomly drawn communities were dropped from the program. The remaining 365 were scheduled to be surveyed. In each community, 10 households were selected at random to field the survey in both rounds, in addition to the village chief. However, we were unable to collect ex-ante and ex-post survey data in every community in our sample. In particular, for 27 out of the 365 targeted communities ex-ante surveys are missing, while ex-post data is missing for 25 communities. This brings our analytical sample for the baseline analysis (using ex-post data) to 340 communities, and 3,372 households.

Besides the addition of the conflict component, a distinctive feature of STEP's CDD program is its *local content*. Whereas most CDDs so far have invested in the creation of new bodies of decision-making, STEP's CDD sought to work with existing institutions (as in Casey et al., 2012), both for the management of the infrastructure work and the conflict mitigation treatment. Doing so avoids duplicating power bodies, bypassing actual centers of power (which can be hard to shift), and provides better guarantees for making any positive institutional impact more sustainable, given that the existing institutions are already part of the broader institutional framework and will continue to exist after the program (see for instance Beath et al. (2013).

This *local content* principle was also applied to the choice of implementing partners. Rather than working with a large international NGO (see e.g., Humphreys et al., 2019), STEP was set up in collaboration with FSDRC and more than 50 local NGOs as implementing partners. The FSDRC was responsible for contracting the NGOs, which were in their turn charged with providing trainings in both participatory decision-making and conflict mitigation, and with helping the LDCs to manage the infrastructure works. The upside of working with local implementing partners is their greater ability to tailor the intervention to the specific local context, as well as the larger external validity for upscaling the experiment as part of a structural government policy (independently from donor involvement). To promote the former, the NGOs received much leeway in the choice of activities to promote participatory decision-making and strengthen conflict mitigation institutions.

Moving on to our findings, beginning with program implementation and community participation in the process, we find that most infrastructure funded projects were implemented on time. The levels of community knowledge of or engagement with the process was however relatively low and uneven across different subgroups. Those who were aware of the STEP project expressed several grievances. Specifically, our data revealed that the infrastructure works finished within the expected time in 82 percent of our study communities. The data revealed however that four out of ten respondents were unaware of the STEP infrastructure project, and only two respondents (<1%) acknowledged that the community received aid in the form of a conflict mitigation program. Among respondents who were aware of the infrastructure project many expressed grievances with the process, which they found tedious and lacking in information-sharing.

Concerning actual program impacts of the CDD and CDD+ interventions, we find no evidence of improved (access to) infrastructure, neither of a strengthening of community cohesion. Specifically, regarding the:

Impact on infrastructure: we do not observe any positive effects on improved infrastructure access and use in the CDD (+) communities. To the contrary, we observe negative effects on most infrastructure outcome variables, whilst mostly insignificant. These null results may be explained by several factors, including delayed and incomplete construction works, as well as inadequate staffing, equipment, and high service fees of the service facilities, which ultimately prevented utilization. Another reason for the null result may relate to the large variety of projects, across six different sectors, and the inability of a single survey instrument to pick up the diverse impacts of e.g., building a market and rehabilitating a school.

Impact on Social Cohesion: we observe few positive effects on social cohesion related outcomes. For instance, we see a decrease in the number of committees organized in the community and attended by household, in both CDD and CDD+ communities. Effects on individual outcome variables confirm this trend. Furthermore, we document an increase in cleavages between different groups, especially in villages that were assigned to the CDD+ intervention. While we observe a slight reduction in the number of conflicts in both treatment arms (and significantly more so in villages that received the basic CDD intervention), conflicts were less likely to be solved at the local level. The source of these effects is not clear. A possible explanation, to be investigated further, could be the unintended consequence of the CDD+ intervention, which may have increased people's awareness and reporting of cleavages and divisions in recipient villages during the implementation process. After all, this entailed bringing the community together to discuss conflicts and provide tools to handle them better.

Impact on socioeconomic wellbeing: we observe null effects for all three outcome families, i.e., economic welfare, income generating activities, and subjective well-being. As these are secondary outcomes, i.e., hypothesized knock-on effects following the primary outcomes, this should not surprise as we find mostly null results on the primary outcomes.

Heterogeneous impact: we find no heterogeneity with respect to project sector and gender but find a few CDD impact outcomes to increase with household wealth. Regarding the project sector, we focus on our main outcome dimensions in the *education* and *health* sectors, which represent 57 percent and 22 percent of all projects, respectively. We fail to detect any positive effects of the education- or health-related CDDs on the overall index of education or health infrastructure, although the individual outcome variables that measure the school's roof and wall quality turn up significant. We also find few significant differences in the results disaggregated by gender. The pre-intervention wealth level generates however differential impacts of the program. Our results on individual outcome variables show that, compared to the control group, wealthier groups who received the intervention had significantly more earnings, were more likely to use the secondary education infrastructure, and were also more likely to be satisfied with the funded infrastructure, though they were less likely to participate in village committees.

In sum, while the local-content-and-leeway principle underlying the STEP CDD interventions evaluated in this study could have strengthened the policy design and its implementation, we find no evidence of improved (access to) infrastructure, neither of a strengthening of community cohesion. In the conclusion of this report, we situate our largely null results within the broader CDD literature, discuss potential explanations for the lack of results, as well as recommendations for policy and future research.

Acknowledgments

This report is the result of a fruitful collaboration between many stakeholders who participated in the design and successful implementation of the STEP impact evaluations: Social Fund of the DRC (FSRDC), the project implementing agency, the Ministry of Social Affairs of the DRC, the Development Impact Evaluation (DIME) Department, the DRC Country Management Unit (CMU) and various Global Practices (GPs) and Cross-Cutting Solutions Areas (CCSAs), in particular the FCV and Gender CCSAs.

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Some illustrative photos of the recruitment process and the work in this report were taken by the Webmaster of the FSRDC, to whom we say a big thank you.

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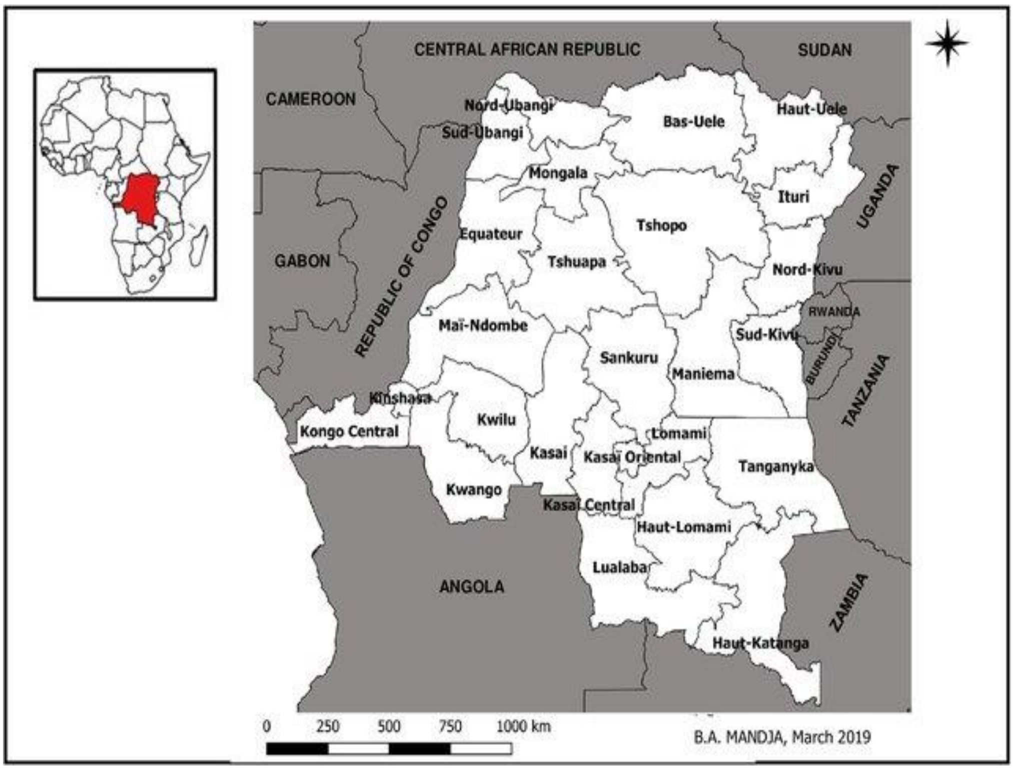
1. Political, social, and economic context

The CDD program took place in six provinces: South Kivu, North Kivu, Haut-Uele, Bas-Uele, Ituri, and Tshopo (see **Error! Reference source not found.**). The latter four provinces formed together Province Orientale, prior to the 2015 administrative partitioning. Combined, the six provinces span approximately 630,000 km², and have a population of about 40 million, thus covering 27 percent of DR Congo's territory and accounting for about 40 percent of its population.[1]

FIGURE 1: DRC PROVINCES



Yet, the six provinces account for 80 percent[2] of all violent events reported in DR Congo since the 2003 peace agreement[3], a period that earned the label 'violent peace'. During the implementation period of the CDD program (2016-2020), more than 120 armed groups operated in the region, and violence was on the rise (Kivu Security Tracker, 2020 (KST, 2020)). In the same period, and in the six provinces combined, 2,805 battles between armed groups and 2,712 acts of violence against civilians were reported, or an average of 47 and 45 per

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[1] Authors' calculations based on information provided by the Congolese Cellule d'Analyses des Indicateurs de Développement (Congolese Cellule d'Analyses des Indicateurs de Développement n.d.) (last consulted on 15/04/2021).

[2] Authors' calculation based on ACLED (2020) (ACLED 2020) last consulted on 15/04/2021.

[3] The 2003 peace agreement put a formal end to the 'second' (1998—2003) Congolese War (preceded by the first Congolese far in 1996-97), which involved eight African nations and 25 armed groups, and has been called the deadliest war in modern African history (IRC, 2007 (IRC 2007))

month, respectively[4]. North-Kivu was most affected, accounting for half of these conflict events. Largely because of this 'violent peace' DR Congo counted 5.5 million internally displaced in 2019 (IDMC, 2020 (IDMC 2020)), 4.5 million of which were residing in the Kivus and Ituri (UNHCR, 2020 (UNHCR 2020)).

While some of the armed groups have foreign roots (M23, LRA, FDLR, ADF, …) most of the groups are local, and many have less than 100 members (KST, 2020). Groups regularly split and dissolve while new groups are created, and group alliances continuously shift (KST, 2021). The protracted conflict and violence is fed by an amalgam of motivations, going from old but unresolved grassroots conflicts over land and local leadership, to more recent conflicts over the control of mining sites, as well as trade routes and other tax bases that provide armed groups with the revenue streams needed to sustain them (Autesserre 2010; KST, 2021; Stoop, Verpoorten and Van der Windt, 2019; Stoop and Verpoorten, 2020).

The conflicts oppose ethnic groups, but also play within clans and even within families, as well as between local communities and multinational companies. Not all conflicts lead to the formation of armed groups and violence, but the division and mistrust they create is detrimental to inter- and intra-community cooperation, thereby hampering development through the under-provision of public goods and services, and the inefficient management of common pool resources (STEP 2015).

Delayed presidential elections further destabilized the period under study. The elections to replace Joseph Kabila, in power since 2001, were originally scheduled to be held in 2016, but took place only in December 2018. The delay caused protests which were violently cracked down on. In the face of mounting domestic and external pressures, Kabila refrained from unlawfully running for another term in the December 2018 elections, and instead put a bet on interior minister Emmanuel Ramazani Shadary as his hand-picked successor. Despite receiving help in the form of voter intimidation and the blocking of some high-profile candidates, Shadary's results were disappointing. After claims of elections irregularities and days of uncertainty about elections results, Félix Tshisekedi, one of the candidates from the opposition parties, was declared the victor of the December 2018 elections (Salihu, 2019).[5] Adding to the perception of fraudulent elections was the suspension of voting in Beni, Beni city and Butembo city, all opposition strongholds. The suspension, conveniently motivated because of 'the persistence of the Ebola epidemic' and 'the terrorist threat' (CENI, 2018), led to street protests in several Congolese cities.[6]

The political uncertainty took place in an already unfavorable macro-economic climate. At the start of the study period, in 2016, GDP per capita growth turned negative (at -0.8%) in the wake of slumping copper and cobalt prices (IMF 2019). In the three-year period that followed

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[4] Authors' calculation based on ACLED (2020) (ACLED 2020) last consulted on 15/04/2021.

[5] https://freedomhouse.org/country/democratic-republic-congo/freedom-world/2020

[6] https://www.hrw.org/news/2019/01/05/dr-congo-voter-suppression-violence

(2017-2019) annual GDP per capita growth was just 1.3% on average, sharply contrasting with the three-year average of 4.8% prior to the downfall (2013-2015) (World Bank 2020). The economic downturn affected the Congolese population mainly through (officious) public expenditure cuts, currency depreciation and inflation (Cassimon, Verbeke and Verpoorten 2016). In the period 2015-2017, for instance, the Congolese franc lost 71 percent of its value and inflation rose by 42 percent (IMF 2019).

The Ebola outbreak further added to economic and social fragility in the period under study. The outbreak was declared in August 2018, and over the course of two years, spread across 29 health zones, including nineteen in North Kivu, nine in Ituri, and one in South Kivu. Most cases were recorded in and around the cities of Beni and Butembo. In total 3,470 cases were confirmed, causing the death of 2,287 people (MSF 2020), which made this tenth Ebola outbreak in DRC the second largest in the world, only surpassed by the 2013-16 outbreak in West Africa. In a study of the economic impact of Ebola in Goma, 20% of respondents indicated that the Ebola outbreak led to an income decrease, 16% said it increased food insecurity, while 18% linked it to a higher incidence of criminality (Stoop, Desbureaux, et al. 2020).

When it comes to internationally comparable development indicators, DR Congo features at the bottom of most rankings. It is estimated that, in 2012, 77% of Congolese were living below the international $1.9 PPP extreme poverty line (WDI, 2020)[7]. More recent poverty estimates are unavailable. The 2019 Worldwide Governance Indicators (WGI 2019), that rate countries against six dimensions of governance[8], put DRC among the bottom 5%. Freedom house rated DRC's democracy a 15 out of 100 in 2019, down from 25 out of 100 in 2015 (freedom house, 2020 (Freedom house 2020)). In the 2019 ease of Doing Business Indicators (WB, World Bank - Ease of doing business ranking. 2019), DRC is among the 5 lowest-ranked countries. On the 2017 Human Development Index (HDI 2017), DRC occupies place 179 (out of 189). On the 2018 Gender Inequality Index (UNDP 2018) it occupies rank 156 (out of 162 countries ranked). Finally, DRC has one of the youngest populations in the world: 65% is below 25 years old (UN, 2019 (UN 2019)).

As indicated at the start of this section, the six provinces under study account for a disproportionally high share of violence within DR Congo. Nevertheless, according to the latest representative survey (1-2-3 survey in 2012), poverty rates are lower the country

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[7] 1.9 USD purchasing power corresponds to 1.2 USD in DR Congo (WB, World Bank 2020). The national extreme poverty line was set at 438,000 CDF, or 1.3 USD a day, and 27% was reported to live below it. The national poverty line was 794,000 CDF or 2.4 USD, and 64% were reported living below it in 2012 (World Bank, 2016 (WB, World Bank - République démocratique du Congo RDC – Évaluation de la pauvreté. Rapport (No. ACS19045). 2016)). Note that the international and national extreme poverty line are very close to each other (1.2 USD vs 1.3 USD) while the reported proportion living below it is very far apart (77% vs 27%). It is beyond the scope of this paper to further dig into this inconsistency.

[8] (1) Voice and Accountability, (2) Political Stability and Absence of Violence, (3) Government Effectiveness, (4) Regulatory Quality, (5) Rule of Law and (6) Control of Corruption

average of 77%, varying in between 49% to 63%. This is due to a relative improvement in security since 2005, leading to a decrease of poverty by almost 20 percentage points in the period 2005-12, combined with fertile soils, a booming mineral sector, and the presence of many international organizations (World Bank, 2016 (WB, 2016). Yet, the situation remains fragile, and the political transition has not yet been able to bring about a turnaround, despite president Tshisekedi vowing to put an end to the insecurity in Eastern Congo.[9]

It is against this backdrop that the international community has been involved in efforts to end conflict and to support economic recovery in Eastern DRC. The World Bank has contributed through the IDA-funded 'Productive Opportunities for Stabilization and Recovery in the DRC' program, or STEP, in its French acronym. STEP is an $80 million project implemented by the Social Fund of the DRC (FSRDC) – a unique arm of the DRC presidency set up for development. Its main stated aim is to improve resilience and livelihoods in conflict-affected communities in North Kivu, South Kivu, and Oriental Province in Eastern Congo. Besides the CDD, with an envelope of $30 million, STEP also included a workfare program, of which the urban component has been evaluated and analyzed in Brandily-Snyers et al. (2021).

2. The Intervention and Theorized Effects

2.1 The basic CDD

The CDD intervention provides communities with a envelop of up to $100,000 for the construction and rehabilitation of community infrastructure, as well as trainings to manage the infrastructure project in an inclusive and participatory way. Communities could select projects from the following categories: health, education, water and sanitation, trade (markets, storage of agricultural products), rural transport (small bridges), energy, and protection of the environment (**Error! Reference source not found.**). Given an average community size of 16,250, this amounts to $6 per person; a budget of which at least three quarters is spent on the infrastructure project, and only a small share on the trainings and assistance.

To enhance inclusive and participatory community involvement in the project's management, the FSDRC organized training and assistance after the disbursements of funds. To further promote community ownership, each recipient community was expected to co-contribute 10 percent of the project envelope, in cash or in kind (labor, local construction material).[10] To implement the CDD program the FSRDC recruited local NGOs and construction agencies that worked directly with the communities and their representatives.

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[9] At the time of writing this report, decreed the state of siege in the provinces of North Kivu and Ituri, from May 6 for a period of 30 days. In an attempt to quell the violence, the provincial and territorial authorities in these provinces were replaced by military ones, and civil jurisdictions will be substituted by military ones.

[10] In exceptional cases, the Social Fund could reduce or cancel the required contribution so as not to hamper the access of the poorest communities

In contrast to many other CDD programs, no new community development structures were created. Instead, FSRDC and local NGOs worked together with already existing development committees, i.e., Local Development Committees (LDCs). The LDCs had to select the construction company, monitor project implementation, and set up a structure for project maintenance and the collection of user fees where possible. Moreover, the LDCs were responsible for the accounting and financial management of the project funds. To ensure participation of the whole community, the LDC and the structure it set up for project maintenance had to include at least a 30% representation of the most vulnerable, including women, internally displaced and youth at risk.

FIGURE 2:THE BASIC CDD PROJECTS

School Rehabilitated by FSRDC in a STEP treatment village

Markets renovated by FSRDC in a STEP treatment village

2.2 Conflict mitigation

A subset of CDD recipient communities received a conflict prevention and resolution add-on intervention seeking to address local conflict and violence. The Eastern DRC context – a context that mirrors many other conflict-affected societies – requires that risks of conflict at the local level be taken into consideration in the project design despite the urgent need of improving basic infrastructures.[11] To implement this component the FSRDC contracted NGOs specialized in conflict prevention and management. The NGOs drew up a conflict mapping for each selected community, and then identified appropriate activities in the areas of conflict mediation and resolution efforts, including for instance the identification and training of key stakeholders in conflict assessment and management, the development or strengthening of internal community dialogue, and the improved reception of returning IDPs.

These activities operated at multiple levels if necessary. At first instance, existing local conflict resolution structures and mechanisms were used, to the extent that due diligence had demonstrated their legitimacy. If necessary, however, the conflict mitigation process could evolve to higher levels if the conflict could not be resolved at the village level.

FIGURE 3: CONFLICT MITIGATION AND GBV PRESENTATION MEETING

2.3 Theorized effects

CDD programs are a popular model for providing economic infrastructure. In addition, through their inclusive and participatory community-driven approach towards the realization of the infrastructure project, they also seek to improve social cohesion.[12] The twin goals of infrastructure and social cohesion are especially appealing in countries experiencing or recovering from civil war, because it is commonly assumed that violence has led to a deterioration of both physical and social capital (King and Samii 2014; Wong and Guggenheim 2018). At the same time, it is in these fragile and conflict-affected contexts that injecting additional resources into communities may exacerbate existing tensions leading to more conflict and social division, or the already existing internal divisions may work against the effective implementation of the project (Wong and Guggenheim 2018). Therefore, our CDD

[11] A survey conducted in 2012 in three hundred villages of South Kivu finds that over 25% of the village chiefs reported to have intervened in one or more disputes between village inhabitants in the month preceding the survey. One major reason for these conflicts was disputes over land (Humphreys et al, 2012).

[12] Enhanced economic and social infrastructure can contribute to community resilience – the overall goal of the Eastern DRC Recovery Project - defined as the ability of communities to respond to disturbances (Matarrita-Cascante, et al. 2017).

augments the classic CDD approach with activities to strengthen local conflict prevention and resolution mechanisms.

The theory of change is illustrated in Figure 4. The first component is the construction and rehabilitation of community infrastructure. The second component of the project, and central to the CDD approach, is the process of participatory and inclusive implementation. The third component is the conflict mitigation component. The different components are theorized to strengthen each other and contribute to the twin goals of improved (access to) economic infrastructure and social cohesion both directly and indirectly.

Many communities in Eastern Congo lack basic infrastructure such as schools, health facilities and paved roads, due to outright destruction by the conflict or a persistent lack of investments. The (re)construction of infrastructure should have a direct impact on the presence of infrastructure. Furthermore, the emphasis on inclusive community participation processes may indirectly contribute to improved infrastructural outcomes by increasing the quality of the implementation and the level of ownership that community members feel they have over the project. This proposition is based on the idea that participatory approaches to development yield better results than traditional top-down approaches (e.g., Scott 1998). By including the voices of local beneficiaries, the distance between principal and agent would decrease, which is likely to produce choices that better reflect their needs (Mansuri and Rao 2012). Finally, insofar as local divisions undermine successful project implementation, the conflict mitigation component also indirectly contributes to improvements in access to infrastructure.

The building of infrastructure through an inclusive community participation process also serves as a vehicle for the second stated outcome, i.e., improving social cohesion. In so far, the CDD program is successful in improving the infrastructure and accessibility to it for everybody in the community, it can result in more acceptance and trust among individuals within the community; decrease the extent to which villages are divided along social, economic, or other lines; and increase the propensities to work collectively within the community to address development challenges. The inclusive process can also increase the satisfaction with and access to the infrastructure works by vulnerable groups (such as women, the poor, the illiterate and ethnic minorities) whose voices may otherwise have been considered less.

According to the ToC the improved social cohesion would persist over time. The mechanism behind this idea is that the CDD project can marshal a type of demonstration effect: experience with working cooperatively with all members of the community for a limited period leads to the adoption of similar practices also outside of the CDD project. Optimistic as it may sound, this idea underlies a large class of development aid projects including many of the largest interventions in post-conflict areas (Mansuri and Rao 2012). At the same time, however, fragile, and conflict-affected communities may be less well equipped to implement the infrastructure project successfully and cooperatively. Moreover, the injection of additional resources may exacerbate existing tensions or create new ones. Therefore, the conflict mitigation component seeks to strengthen local conflict prevention and resolution mechanisms, and by doing so enhance the twin goals of improving access to infrastructure and social cohesion.

FIGURE 4. THEORY OF CHANGE

Aim	**Activities**	**Outputs**	**Outcomes**
Improving access to community infrastructure	- Build or rehabilitate infrastructure by local contractors, including equipment where appropriate (e.g., desks for schools) - Technical assistance and supervision throughout project implementation to ensure high technical quality.	Infrastructure has been built	
Facilitating and improving inclusive community participation processes	- FSRDC conducts a community sensitization campaign to share information about the project - FSRDC organize and train community members to prioritize and select community infrastructure needs and develop a project proposal - FSRDC pre-qualifies and trains local NGOS and contractors - Communities identify an NGO (pre-qualified by FSRDC) - After fund distribution, training and assistance will be provided to ensure proper management of funds by the communities, monitor project implementation and organize maintenance	- Ownership by the community of the project - Inclusion of the most vulnerable	1. Improved access to infrastructure 2. Improved social cohesion
Strengthening local conflict prevention and resolution mechanisms	- In-depth and ongoing conflict-sensitivity analysis - Mediation - Conflict management training and support	Resolution of disputes	

In sum, we hypothesize:

- ◊ **H1**: The community-driven development program leads to improvements in the quality, access and use of community infrastructure.
- ◊ **H2**: The community-driven development program leads to improvements in social cohesion.

- ◊ **H3**: Communities that implement CDD projects with an explicit conflict mediation/resolution mechanism should see more improvement in access to community infrastructure and social cohesion than communities that receive CDD projects without conflict resolution mechanisms.

Secondary hypotheses relate to the heterogeneities of the impact. Since the CDD process aims to be inclusive, we also expect a high satisfaction of vulnerable groups with both the process and the infrastructure works. Concretely, this implies the following hypothesis:

- ◊ **H4**: Vulnerable groups have improved access to and use of community infrastructure.
- ◊ **H5:** Vulnerable groups experience an improvement in social cohesion

Vulnerable groups could be women, the illiterate, ethnic minority groups, and/or the poorest, i.e., those who usually weigh less in decision making.

3. Experimental design and empirical strategy

3.1 Assignment to the treatment

Before the start of the CDD project, FSRDC conducted a community sensitization campaign throughout the six provinces to share information about the program and organize and train communities to prioritize and select community infrastructure needs and develop a project proposal. In the months after this sensitization campaign, the FSRDC received the project proposals and judged their quality. Only those that passed a set of predetermined criteria were eligible for the CDD project.[13] The number of approved projects for grants was significantly larger than the number of projects that could be financed given the resources at the disposal of FSRDC. This allowed us to randomly assign approved projects to the CDD intervention, thereby creating a control and treatment group that are similar in expectation.

The CDD program was implemented from 2016 to 2020, with a new enrolment period at the beginning of each year. To keep randomization logistically feasible, all provincial FSRDC headquarters received a 'randomization list'. New communities were added to this list as they came in (row 1, row 2, row3, etc.). Every subsequent three rows on the list were assigned

[13] The following criteria were taken into account: projects that (1) improve living conditions and cohesion, peaceful coexistence and inclusion of different community groups; (2) are technically and socio-economically viable, without high environmental or social negative externalities; (3) can be implemented within a period not exceeding 12 months; (4) rely on a maximum of local resources, both human and material; (5) primarily benefit vulnerable population groups; (6) do not require more than 100,000USD for implementation; (7) are complementarity with the activities carried out by other actors. There were no restrictions on what constitutes a 'community'. The unit of implementation are typically small, natural entities (e.g., villages or localities), but neighbourhoods within a village, or the head of the province, could also decide to apply for a project. For the duration of the impact evaluation, we will refer to the units under study with the generic term 'communities.

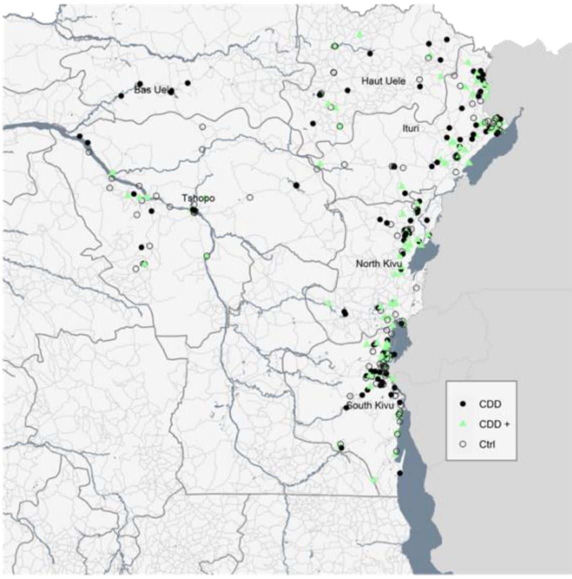
randomly to CONTROL, CDD or CDD+. In total, 400 villages were assigned across these three categories. Table 1 shows their distribution across CONTROL, CDD or CDD+.[14]

TABLE 1. RANDOMIZATION OF VILLAGES PER TREATMENT ARM

Control communities		CDD Treatment Communities	
		No conflict mitigation	Conflict mitigation
<u>A:</u> 127 communities		<u>B:</u> 138 communities	<u>C:</u> 135 communities

Note: The Table presents the number of communities assigned to control and each of the two treatment arms, over the period 2016 to 2019

Figure 5. Location of CDD, CDD+ and control communities



3.2 Measurement of outcomes

We focus on measuring the program effects along two main dimensions: infrastructure and social cohesion. We divide each of them in several families of outcomes. Table 2 gives an overview of these outcome families; Table A 4 in Appendix 2 provides the measures included

[14] The imbalance of communities across the three experimental groups relates to the phase-in design and specificities of the randomization: from 2016 to 2019, four randomizations occurred, i.e. in each calendar year (over the 6 provinces). Moreover, the randomization was stratified by project type, to make sure we would have an even number of project type by territories.

in each family, as well as a careful mapping between these measures and the variables collected through the surveys.

TABLE 2. OUTCOME FAMILIES

<table>
<tr><th>Theory of change: Aim</th><th>Main outcome domain</th><th>Outcome family</th></tr>
<tr><td colspan="3">Primary outcomes</td></tr>
<tr><td rowspan="6">Improving access to community social and economic infrastructure</td><td rowspan="6">Access to and quality of socioeconomic Infrastructure</td><td>Infrastructure access (HH & Village level)</td></tr>
<tr><td>Infrastructure use (HH level)</td></tr>
<tr><td>Health Infrastructure (HH level)</td></tr>
<tr><td>Education Infrastructure (HH level)</td></tr>
<tr><td>Satisfaction with STEP (infrastructure) (HH level)</td></tr>
<tr><td>Infrastructure provision (village level)</td></tr>
<tr><td rowspan="5">Facilitating and improving inclusive community participation processes</td><td rowspan="8">Social cohesion</td><td>Community organization (within village)</td></tr>
<tr><td>Information transmission (within village)</td></tr>
<tr><td>Trust (within village and between villages)</td></tr>
<tr><td>Ethnic division/cleavages</td></tr>
<tr><td>Social cohesion</td></tr>
<tr><td rowspan="3">Strengthening local conflict prevention and resolution mechanisms</td><td>Conflict & disputes (within village)</td></tr>
<tr><td>Conflict & disputes (between villages)</td></tr>
<tr><td>Inclusion of the most vulnerable (within village)</td></tr>
<tr><td colspan="3">Secondary outcomes</td></tr>
<tr><td rowspan="3">Socio-economic well-being</td><td>Economic Welfare</td><td>HH assets ownership (HH level)
HH consumption expenditures (HH level)</td></tr>
<tr><td>Income Generating Activities</td><td>Employment (HH level)
Working Hours (HH level)
Earnings (HH level)</td></tr>
<tr><td>Subjective Well-Being</td><td>Self-Perception of life conditions (HH level)</td></tr>
</table>

Regarding infrastructure access and quality, we measure the quantity and quality of infrastructure, to what extent individuals have access to this infrastructure, are satisfied with it, and whether it has led to improvements in outcomes such as health and education. At the village level we also study the effect of the intervention on the provision of infrastructure by different actors (village member, government, NGO).

To capture social cohesion, we constructed eight different families. The family 'community organization' includes measures on the existence, participation in, and meeting frequency of village committees. The second family, 'information transmission', captures to what extent information available to the village chief is known to households in the village. Family three measures intra- and inter- group trust. The fourth family, 'cleavages' captures within-village divisions between poor and rich, young and old, rich and poor, etc. Family five, labelled 'cohesion', measures attitudes of households towards returned internally displaced, returned refugees and individuals of other ethnic groups. Family six and seven treat with within-village and between-village conflicts, respectively, including information on the frequency and type of conflicts, as well as the nature (violent or not) and whether conflicts were resolved, and by whom (the chief, another local actor, or an external actor). Finally, the family 'inclusion' measures attitudes towards internally displaced, refugees, ex-combatants, and neighboring villagers.

Our secondary outcomes aim to capture the effect of the intervention on the economic life of the participants. They include variables characterizing socioeconomic well-being, namely household asset ownership, household consumption expenditures, employment, working hours, earnings, and self-Perception of life conditions.

To evaluate the degree of compliance, we perform a set of manipulation checks. In particular, we expect that the treatment group should be significantly more aware and more involved in the STEP infrastructure project. To verify this, we rely on nine variables related to this project and are summarized in Table 3 and further detailed in Appendix Table A 5.

TABLE 3. MANIPULATION CHECK VARIABLES

Manipulation Check	
Knowledge of STEP	Do you know about the STEP project?
STEP was built	Was the project built
Satisfied with STEP	Summation across 18 different categories on satisfaction
STEP use by respondent	Last week, how many times did you personally use the STEP infrastructure?
STEP use by household	Last week, how many times did you personally or your household members use the STEP infrastructure
Project Identification	Participated in identifying needs of community.
Project Contribution	Participated in project contribution mobilization.
Project Execution	Participated in project execution.
Project Administration	Participated in project administration.

Some variables were removed from the analysis, for instance because there was almost no variation in the responses. We carefully catalogued any changes from the pre-analysis plan in Appendix 3.

3.3 Data collection

We measured the outcomes of interest through two data collection rounds: an ex-ante and an ex-post round. The surveys were implemented in each community on a rolling basis, shortly before the start of the CDD project, and on average eleven months after the project finished.[15] In each community, in addition to the village chief, 10 households were selected at random to field the survey in both rounds. The village chief survey collected information related to community characteristics such as the presence of community infrastructure, but also information about divisions in the community, disputes that took place preceding the survey and the actions by the chief to overcome them. The household survey largely focused on information related to household and individual level characteristics. It included questions about access to infrastructure, participation in community events, perceived divisions, altruism, and trust towards fellow villagers, etc. Similar data collection exercises took place in control communities.[16]

FIGURE 6:QUANTITATIVE FIELDWORK

FSRDC enumerators undergoing training and implementing the survey in the field.

[15] Baseline surveys were collected in between September 2016 and December 2019; endline surveys in between July 2017 and December 2020. The time in between the end of the construction and the ex-post survey varies between 4 month and a half and 27 months and half.

[16] Data was collected using electronic tablets. Surveys were programmed in Open Data Kit and automatically uploaded and stored online via the surveyCTO platform. To ensure reliability, back checks took place for at least 5% of the collected data. The surveys built on other surveys related to similar impact evaluations to maximize opportunities for learning. For example, the survey was built on the 2012 survey conducted by Humphreys et al. (2012) (Humphreys, De La Sierra and Van der Windt, Social and Economic Impacts of Tuungane Final Report on the Effects of a Community Driven Reconstruction Program in Eastern Democratic REPUBLIC of Congo. 2012) in Eastern Congo that had as goal to learn about a CDD project implemented between 2007 and 2012.

3.4 Estimation strategy

We estimate the intent-to-treat (ITT) effects of the intervention, using the following linear regression:

$$Y_{ihv} = \beta_0 + \beta_1 * CDD_v + \beta_2 * CDDM_v + \gamma X_{ihv} + \epsilon_{ihv} \qquad (1)$$

where Y_{ihv} is the outcome of interest for individual *i* in household *h* in village *v* as measured in the ex-post survey; CDD_v is a dummy variable indicating assignment of the community to the CDD program excluding the conflict mitigation component; $CDDM_v$ is a dummy variable indicating assignment of the community to the CDD+ program; X_{ihv} is a vector of covariates, and indicators for randomization strata (province-by-cohort/enrolment-year and sector of the project); ϵ_{ihv} is an idiosyncratic error term. To account for dependence between respondents within the same community, we cluster ϵ_{iv} at the village level. For outcome measures at the community level rather than the individual level, we replace y_{iv} with y_v, and use simply robust disturbance terms. Following hypotheses 1 and 2, we expect: β_1>0. In addition, following hypothesis 3, we expect (β_2-β_1)>0.

For communities for which we have both ex-ante and ex-post survey data, we additionally use a difference in difference (DiD) framework.[17] The DiD approach allow us to compare the average change in outcomes in treated and control communities by taking into account the baseline level of each outcome studied. We use the following DiD specification:

$$Y_{ihvt} = \beta_0 + \beta_1 * CDD_v + \beta_2 * Post + \beta_3 * CDD_v * Post + \beta_4 * CDDM_v + \beta_5 * CDDM_v * Post + \gamma X_{ihvt} + \epsilon_{ihvt}$$

where Y_{ihv} is the outcome of interest for individual *i* in household *h* in village *v* at time t (ex-ante or ex-post survey), CDD_v is a dummy variable which indicates assignment to the treatment group, and controls for baseline imbalances across groups, $Post$ is a dummy variable as well, indicating the post-treatment period. The variable of interest, $CDD_v * Post$, is an interaction of time and group assignment dummies and isolates the treatment effect on the treated group after treatment took place. CDDM is a dummy variable that relates to the conflict mediation treatment assignment, and $CDDM_v * Post$ is an interaction of time and conflict-mediation treatment assignment which isolates the effect of the conflict mitigation treatment on the treated group after treatment.X_{ihvt} is a vector of (ex-ante and ex-post) covariates, and indicators for randomization strata (province-by-cohort/enrolment-year and sector of the project); ϵ_{ihvt} is an idiosyncratic error term.

[17] Due to time constraints, the current draft report only includes ITT results. DiD results will be included in future versions.

In addition to reporting the effects on each outcome of interest, we estimate the effects on standardized indices of outcomes part of the same family. To construct the indices, we follow the approach pioneered by Kling, Liebman and Katz (2007). [18]

For communities for which we have both ex-ante and ex-post survey data, we additionally use a difference in difference (DiD) framework. The DiD approach allow us to compare the average change in outcomes in treated and control communities by taking into account the baseline level of each outcome studied. We use the following DiD specification:

$$Y_{ihvt} = \beta_0 + \beta_1 * CDD_v + \beta_2 * Post + \beta_3 * CDD_v * Post + \beta_4 * CDDM_v + \beta_5 * CDDM_v * Post + \gamma X_{ihvt} + \epsilon_{ihvt}$$

where Y_{ihv} is the outcome of interest for individual *i* in household *h* in village *v* at time t (ex-ante or ex-post survey), CDD_v is a dummy variable which indicates assignment to the treatment group, and controls for baseline imbalances across groups, $Post$ is a dummy variable as well, indicating the post-treatment period. The variable of interest, $CDD_v * Post$, is an interaction of time and group assignment dummies and isolates the treatment effect on the treated group after treatment took place. CDDM is a dummy variable that relates to the conflict mediation treatment assignment, and $CDDM_v * Post$ is an interaction of time and conflict-mediation treatment assignment which isolates the effect of the conflict mitigation treatment on the treated group after treatment.X_{ihvt} is a vector of (ex-ante and ex-post) covariates, and indicators for randomization strata (province-by cohort/enrolment-year and sector of the project); ϵ_{ihvt} is an idiosyncratic error term.

3.5 Implementation timeline

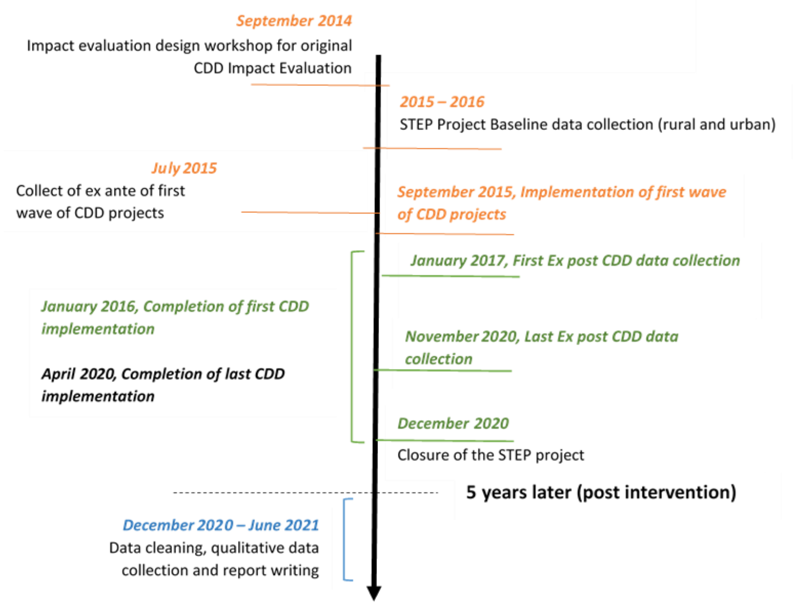
This Impact evaluation includes three phases that ran over the course of 5 years: design, implementation, and analysis. The design phase began with an IE workshop, which took place in Goma, DRC, in September 2014 and included the FSRDC team, provincial government representatives, representatives from other international organizations and NGOs operating in social protection and community development sectors and the DIME team. The initial design concept from this workshop was further developed and the IE design was finalized in subsequent months. Implementation began with a project-wide baseline data collection and random sample of villages in targeted provinces between 2015 and 2016.

Ex-ante data collection of the first wave of CDD projects began in July 2015, only two months before the implementation of the associated infrastructure works. The first wave of CDD projects completed implementation in January 2016 and first ex-post data collection began about a year later, in January 2017. Completion of the last CDD project implementation took place in April 2019, while the last ex-post CDD data collection was carried out in November

[18] This is done as follows. First, we redefine each of the variables of interest in a family, so that higher values for each variable imply positive effects. Second, we rescale each of the redefined variables using the (weighted) mean and standard deviation of the control group units. The index is then the standardized average of the redefined rescaled variables.

2020. The STEP project (including the CDD component) effectively closed in December 2020. Data cleaning and analysis has been carried out in the past several months. The IE timeline is summarized in w. below.

FIGURE 7:IMPLEMENTATION TIMELINE



4. Program implementation and community participation

In this section we provide insights in program implementation, as well as the participation of community members, and how the program was appreciated by the beneficiaries.

4.1 Implementation

The infrastructure program was implemented in most CDD and CDD+ communities. In 82% among selected CDD and CDD+ there was compliance: the infrastructure works finished prior to the ex-post survey. In 15% of CDD and CDD+ communities there were considerably delays, to the extent the construction was still to be launched at the time of the survey; and in 3% of communities the construction was dropped due to a lack of funding. Among the control communities, we find 22 (18.6%) cases of communities who started the infrastructure work before the ex-post survey, this despite them being urged to wait for their turn in the roll-out schedule.

4.2 Knowledge and use of STEP CDD

Respondents were asked whether they knew of the STEP construction project. The data reveal that only a small minority (8%) of respondents in the CDD and CDD+ communities recognized the project and the name STEP; 35% knew about the project but not its name 'STEP', and 17% had heard about the infrastructure project, but was unaware that it was part of a development project managed by the LDC. Four out of ten respondents were totally unaware of the infrastructure project. The corresponding percentages for control communities are 5%, 30%, 6% and 58%. Table 4 gives an overview.

TABLE 4. RESPONDENTS' KNOWLEDGE OF STEP CDD

Does the respondent know STEP?	CDD	CDD+	Control
Does not know the project at all	44.64	35.16	57.76
Knows (heard about) the work, but does not know who made it / who ordered i	15.91	18.22	6.36
Knows the project but did not know the name ("step")	30.73	39.19	30.45
Knows what it's about	8.73	7.42	5.43
N	1100	1092	1179

From Table 4, respondents' knowledge of STEP CDD, it is clear that awareness of the project was much greater in treatment communities than in control. The manipulation check (in Table 16) indicates however that only for CDD+ communities, the difference with the control group was statistically different from zero.

The fact that awareness of the STEP project is still relatively large in the control group may be due to information transmission, from treatment to control communities, but it may also result from the fact that the control communities submitted an eligible project and are programmed to receive the intervention at a later stage. Moreover, to identify and design such an eligible project, the control communities also received assistance from an NGO. If this identification and design phase had a persistent impact on decision processes in the community (and social cohesion), then our estimated CDD impact of these outcomes would be biased downward.

Noteworthy is that STEP awareness is lower among two vulnerable groups: women and the illiterate. The poor[19] and ethnic minorities, proxied by those who report speaking a minority language at the community level, are however not less informed about the project. Table 5 gives an overview.

TABLE 5: VULNERABLE GROUPS AND KNOWLEDGE OF STEP CDD

Does the respondent know STEP?	all	female	illiterate	poor	minority language group
Does not know the project at all	46.2	48.9	52.4	47.2	45.9
Knows (heard about) the work, but does not know who made it / who ordered it	13.3	14.9	16.6	14.5	13.6
Knows the project but did not know the name ("step")	33.4	31.6	29.7	32.5	32.5
Knows what it's about	7.2	4.7	1.3	5.9	8.0
N	3371	1309	849	1701	750

Table 6 resents data on the use of STEP. We find that about 20% of respondents in treatment communities reported having used the STEP infrastructure in the past week, compared to

[19] The category 'poor' is defined as the bottom half of the sample in terms of asset wealth, consumption and earnings.

only 1.4% in control communities. This suggests that spillovers from STEP infrastructure use to control communities are moderate.

TABLE 6.USE OF STEP BY RESPONDENT AND HIS/HER HOUSEHOLD MEMBERS

Use of STEP		CDD	CDD+	Ctrl
Respondent used STEP infrastructure in past week		20.2%	20.9%	1.4%
Household member used STEP infrastructure in past week		26.9%	30.6%	4.9%
	N	1101	1092	1179

Notes: in case the respondent did not reply to this question because (s)he did not know the project; the value was set to zero.

Breaking down STEP use by vulnerable groups, we find somewhat lower use among women and ethnic minorities, but not among the illiterate, and even a slightly higher use among the poor.

TABLE 7. VULNERABLE GROUPS AND USE OF STEP INFRASTRUCTURE

Use of STEP	All	Women	Illiterate	Poor	Minority language group
Respondent used STEP infrastructure in past week	14%	12% *	13%	16% ***	11% ***
Household member used STEP infrastructure in past week	20%	18% ***	20%	23% ***	15% ***
N	2193	853	556	1178	446

Notes: *, *, *** indicate a statistically significant difference between the binary categories' women/men, illiterate/literate, poor/non-poor, and minority/majority language group, at 10%, 5% and 1% of a t-test

4.3 Household satisfaction and involvement with STEP CDD

In this section, we focus on the CDD and CDD+ communities. Respondents were asked about 14 potential grievances related to the STEP project.[20] These are summarized in Table 8. Note that the response rate is below 50%: only those sufficiently familiar with the STEP project replied to these questions. Hence, some caution is required when interpreting the results.

[20] The exact survey question was: "Each person may have different concerns or opinions regarding the execution of the STEP projects. I'm going to read you a list of opinions and you tell me if you "Strongly Agree", "Okay", "Disagree", "Strongly Disagree". You can also, of course, not have an opinion (in which case you just have to tell me "

TABLE 8. GRIEVANCES ABOUT STEP: (DIS)AGREEMENT BY RESPONDENT

	In agreement with grievance (%)				
	All	Women	Illiterate	Poor	Minority language group
The whole process (project decision and execution) took too long	51%	52%	48%	47%***	45%*
The ALE (NGO in charge of the project) did not behave well in the village	24%	23%	20%	25%	33%***
The selected projects were not the most important for this village	11%	9%	11%	12%	13%
The selected projects did not help enough people in this village	11%	12%	14%	12%	12%
I had no real influence on the selection of projects	60%	67%***	68%***	57%*	61%
Disagreements about the project in this village were not well managed	15%	14%	11%*	17%	11%
The process required too many steps / formalities to follow	56%	55%	53%	53%	53%
There was not enough information about the process	45%	47%	48%	45%	47%
There has been embezzlement in this village (corruption, nepotism, etc.)	15%	12%	11%*	13%	19%*
The allocation of money across villages was not done fairly	18%	17%	17%	18%	19%
The project created divisions or conflicts in the community	7%	6%	6%	7%	9%
The project created divisions or conflicts between the communities	6%	5%	3%**	6%	7%
The project was controlled by the chief	69%	70%	67%	72%**	70%
The project did not reflect your concerns	11%	11%	11%	12%	11%
Total N	2193	853	556	1178	446
Average response rate	44%	42%	43%	45%	39%

Notes. Based on CDD and CDD+ respondents, *, *, *** indicate a statistically significant difference between the binary category's women/men, illiterate/literate, poor/non-poor, and minority/majority language group, at 10%, 5% and 1% of a t-test

The top three grievances on which respondents agreed most are, in order of importance: 'The project was controlled by the chief '(69%)[21], 'I had no real influence on the selection of projects' (60%), and 'The process required too many steps / formalities to follow' (56%). These are followed by 'The whole process (project decision and execution) took too long' (51%), 'There was not enough information about the process' (45%), and – with a considerably lower frequency 'The ALE (NGO in charge of the project) did not behave well in the village' (24%). There seem to have also been some embezzlement (or perceptions thereof): 'There has been embezzlement in this village (corruption, nepotism, etc.)' is confirmed by 15% of respondents, but at the same time has the lowest response rate, at 36%.

Response rates regarding the grievances were similar among women, the illiterate, and the poor, though somewhat lower among ethnic minorities. Overall, these 'vulnerable' groups express similar opinions about the different grievances, with some exceptions: women and the illiterate agreed considerably more with the grievance 'I had no real influence on the selection of projects'; respondents' part of an ethnic minority were more concerned about the NGO's behavior in the village and about embezzlement of funds; and the poor perceived more control by the chief over the project.

The enumerators also specifically asked respondents in which way they were involved in the STEP project. Table 9 gives the results.

[21] We do not think that all respondents who confirm this statement find it problematic that the chief is in control. To the contrary, the expectation is that a chief should be strong and therefore should be in control. This interpretation is corroborated by the narratives in response to open questions (see further below).

TABLE 9. RESPONDENTS' INVOLVEMENT IN STEP PROCESS

Implication of respondent in:	CDD	CDD+
Identification phase of community needs	12%	13%
Mobilization of different contributions (helped to finance the project)	13%	16%
Executing the project	16%	21%
Managing the project	7%	9%
N	1101	1092

We find that only 10 to 20 percent of respondents were directly involved in STEP decision making and execution. In the manipulation check (reported in Table 16) these responses do not turn up significantly compared to those in control communities.

We find significantly less participation by vulnerable groups, and this for all four groups, across all four indicators of active participation, with one noteworthy exception: 'execution of the project' by the 'poor'. In case of manual labor, this could be partly driven by self-selection by the poor in low-paying jobs. Hence, it should not per se be seen as a merit of the CDD 'participatory and inclusive' process, certainly given the relatively low scores of the vulnerable groups on all other indicators of participation (Table 10).

TABLE 10. RESPONDENTS' INVOLVEMENT IN STEP PROCESS

Implication of respondent in:	All	Female	Illiterate	Poor	Minority language group
Identification phase of community needs	16%	10%***	8%***	13%***	12%**
Mobilization of different contributions (helped to finance the project)	17%	12%***	12%**	14%***	12%***
Executing the project	22%	16%***	19%**	21%	15%***
Managing the project	10%	6%***	5%***	8%**	6%**
N	2193	853	556	1191	446

Notes: *, *, *** indicate a statistically significant difference between the binary categories' women/men, illiterate/literate, poor/non-poor, and minority/majority language group, at 10%, 5% and 1% of a t-test

4.4. Observations by household respondents

Our survey instrument also allowed for 'narratives' to be constructed from open-ended questions. In this subsection, we review the answer to two open questions:

- What did you not like at all about the realization of the [STEP] project?
- Final thoughts (at the very end of the survey)

To the first question, 413 out of the 2193 CDD (+) respondents expressed a grievance. It is instructive to briefly go through the main grievances that were voiced. They can be divided into grievances related to the construction of the infrastructure (166), to the ex-post management of it (19), and grievances related to the STEP process (224).
The construction-related grievances were voiced by 166 out of the 413 respondents: 73 mentioned a technical defect or the use of materials of inferior quality; 35 would have wished

a larger construction (e.g. more classes to accommodate all pupils); 13 respondents regretted that no fence was built to protect the building, or the pupils; 25 respondents mentioned that no water or sanitary facilities were put in schools, health centers or the market place; 12 lamented the lack of (sufficient) electricity to lighten the building. In a few cases, respondents thought that the location of the construction site was not well chosen. Box 1 provides some illustrations.

Box 1. Construction-related grievances

- ◊ No fence was built to guarantee the safety of students.
- ◊ The school is good, but we need more rooms because the number of pupils has increased following the construction of this new building.
- ◊ They did not dig a water well to help the sick at the health center.
- ◊ The company was not efficient and today almost nothing is completed.
- ◊ The market is too small.
- ◊ The solar panels are too weak to guarantee lighting.
- ◊ The quantity of cement was not respected, and the walls are not strong enough now.
- ◊ There is no road leading to the market.
- ◊ The windows and doors were poorly installed and cannot open.
- ◊ The bridge is too short for the width of the river.
- ◊ There are too few classrooms, and they are not equipped. They should have put light so students could study in the evening.
- ◊ The construction is poorly executed because the water infiltrates the building.

The grievances that relate to the ex-post management of the infrastructure mainly point to access issues, and lack of complementary inputs. As such 15 respondents mention that the health centers and schools are not functional due to lack of staff and equipment, while 4 respondents mention that schools and health centers cannot be accessed by everyone due to the fees charged.

Box 2. Grievances with respect to ex-post management

- ◊ At the start of the project, we were told that the water would be free but today we are required to pay which is why our household no longer uses this structure.
- ◊ The health center has no medication, no doctors, and no camps for nurses.
- ◊ They built a school but did not train teachers.
- ◊ The school is like a paradise, but the children cannot access it because of the high fees.
- ◊ Nothing. Only the room is not used too much.

Among the 224 grievances regarding the STEP-process, 60 respondents referred to the hiring of labor from elsewhere rather than among the village inhabitants and youth; another large group of 48 people thought that the construction took too much time, while an almost equally large number of 46 people mentioned the lack of information and involvement of the entire community in the process. 34 respondents complained about the low, delayed, or even non-payment of the workers or other poor working conditions (e.g., no or bad meals). 14 complained about the bad behavior of the construction firm, the NGO, or a lack of follow-up on the part of the FSDRC; 6 alluded to the embezzlement of funds, and 5 respondents considered the 10% community contribution is much too high.

Box 3. Grievances related to the STEP process

- They NGO brought workers from very far and forgot the youngsters of our community.
- The same young people started and finished the work; instead, they should have rotated.
- The local contribution [of 10%] was exaggerated and led to debts.
- I didn't want this infrastructure work to be done; we will die of starvation.
- There were other priorities than this room: the health facilities.
- I was not contacted for the project; otherwise, I would have told the social fund to build us a maternity.
- Poor awareness because the population was not aware of anything.
- The local chiefs were not involved in the monitoring.
- The workers were not paid; some suppliers were not paid; and construction materials were embezzled.
- There were no lunches of sufficient quantity for the workers.
- The FSDRC did not respect the time limit; the NGO did not really carry out the work wisely; they were cheated by the engineers who sold bags of cement in the village.
- The time scheduled for the work was 4 months but bow we are 12 months ahead and the infrastructure is not yet operational.
- The NGO that is carrying out the project does not behave well in the village.
- The absence of regular visits from the social fund.
- The allocation of contracts to suppliers was not well done.
- Disputes between the NGO, the FSDRC and workers.
- Disputes between some members of the community.

Out of the 2193 CDD (+) respondents, 821 made a final comment at the end of the interview. In 203 cases the respondent thanked the FSDRC for the infrastructure; and in 466 cases the respondent (also) asked for another project or encouraged the FSDRC continue its efforts.

283 respondents made another comment, often repeating one or more of the grievances listed above, but also including some associated recommendations directed at FSDRC.

The following recommendations were made by the respondents: to first make sure that people or not hungry; to recruit labor locally (and fairly, through a lottery); to make sure that schools and health centers are equipped, staffed and accessible at low fees; to carefully select capable NGOs and construction companies; to closely monitor those entities in order to avoid embezzlements; to revise the community's own contribution downward from 10% to 5%; to involve the entire community (including women and minority ethnic groups); to work with the village LDC rather than the 'groupement' LDC, and more closely involve the village chief.

4.5 Observations by village chiefs

Among the 215 chiefs of CDD (+) villages, 42 gave a final remark. These remarks are fully in line with those made by the households. Hence, absent an important discrepancy, we do not further discuss them here.

4.6 Field observations by enumerators

In about 25% of communities, enumerators wrote some observations. Only very few of these observations were neutral (e.g., 'construction work in progress'). In most cases, the observations referred to issues with the implementation of the infrastructure project.

By far the most frequent issue raised (in 30% of the observations) was the low capacity of the construction firm. In more than eight out of ten of these cases, the low capacity was also linked to the financial management by the construction firm, for instance:

Box 4. Grievances by enumerators on the STEP work

- ◊ The weak capacity of the company in the management of funds was at the root of the delay. The technical visit has been organized but the contractor does not consider the reservations formulated therein. A week has been granted to the company to correct the defects; otherwise, the said reservations could be executed by another contractor at the expense of the company.
- ◊ The weak capacity or mismanagement of funds by the company (case of litigation: debts of suppliers and labor). This situation did not allow the project to be completed on time.
- ◊ The mismanagement of funds by the company was at the root of multiple work stoppages, characterized by the non-payment of debts to local suppliers. The company has been criticized for being run by a proxy. The money would first have to pass through Kinshasa when it was disbursed and very little of the funds were allocated to work on the site in return.
- ◊ The weak financial capacity of the company, diverting funds.

The LDCs were tasked with overseeing the work of the construction company, and it was foreseen that FSRDC, and the local NGOs would support the LDCs in carrying out this task, by providing the LDCs with assistance and training. In 10% of the observations made, the follow-up by the NGOs and FSDRC was judged insufficient. For instance:

Box 5. Grievances by LDCs on the STEP management

- ◊ Lack of follow-up by the NGO is at the root of the delay.
- ◊ Delays due to communication deficit between ALE and FSRDC.
- ◊ Weakness in the follow-up by the NGO.
- ◊ Lack of support from the NGO and supervision of the FSDRC Antenna.
- ◊ Poor management of the site by the supervising agency and the NGO.
- ◊ Difficulties with a problematic NGO.
- ◊ Embezzlement by the company and the NGO.
- ◊ Poor management of the site by the company and laissez-faire by the NGO.

Since such observations were not systematically made (the enumerator could leave the space blank), it is difficult to assess how widespread the weakness of follow-up and support was. It is in any case likely that there was considerable heterogeneity, because FSDRC contracted over 50 local NGOs to carry out these tasks, and it is likely that there had different (levels of) capacities.[22]

Another observation that was made frequently (in 15% of observations) related to the difficulty of the communities in mobilizing their 10% contribution to the project (in cash or in kind). For instance:

Box 6. Grievances by enumerators on the communities

- ◊ Difficulty for the community to mobilize the local contribution.
- ◊ Delay in mobilizing the full local contribution by beneficiaries was the major cause of the delay in implementation.
- ◊ Contribution deficit.
- ◊ Problem of mobilizing the contribution of beneficiaries which blocks the progress of the work.

A particularly striking case of deception with respect to the community contribution took place when, during the evaluation of the local contribution, the community had presented a kiln for brick making not far from the construction site. However, later it turned out that the

[22] For 96 out of 435 village chief surveys conducted, the name of the NGO involved was recorded. Among these names, we can distinguish 53 different NGOs, mainly small local ones. Given the many missing observations, it could be that many more NGOs were involved.

kiln did not belong to the community, leading the construction company to halt its operations on several occasions, until the gap in contributions was filled.

Insecurity issues complete the top-5 of most common observations made. For instance:

Box 7. Grievances by enumerators on insecurity issues
◊ The presence of militias in the area causes the withdrawal of key personnel from the company and the Control Office. ◊ Insecurity is at the root of the delay (interethnic conflict between Hema-Lendu in the territory of Djugu). The works have been stopped since the beginning of February 2018 because of this situation. The resumption of work is conditional on the re-establishment of the security situation. ◊ Permanent insecurity in the project area. ◊ Completed after the deadline mainly due to insecurity in the area, this situation lasted for at least 2 months, thus causing the work to be stopped. ◊ Multiple stoppages of the construction site caused by clashes between militias in the area where the project is located.

A few observations mentioned that the projects triggered conflict, e.g., between different interest groups within the village, or because the community opposed against the hiring of skilled workers from elsewhere (despite the sometimes-complicated technical work required). Other observations made related to badly executed constructions, technical defects, cost overruns, lack of construction materials, the difficult access to some production sites, and the destruction of constructions by torrential rains.

Combined with the narratives of the household survey respondents, these field observations suggest that the implementation of the infrastructure projects was not always easy.

4.7 The conflict mitigation and the (local) actors involved

Awareness of the conflict mitigation intervention seems to have been low, both among households and chiefs. The survey probed about development programs that the village received. In particular, the following question was asked "During the past 6 months, has the village received money, food or any other social assistance from the government, NGOs or international donors?". When the respondent said yes, this was followed by a list of options, among which 'infrastructure project' and 'conflict mitigation project'. Only 10% of households mentioned that the village had received some form of aid, and this share is similar across treatment and control**. Only one household in a CDD+ community (and one in a control community) mentioned that the aid came in the form of a conflict mitigation project.** Among the chiefs, about 13% confirmed that the village had received aid, and this share is somewhat higher among chiefs of control communities. **None of the chiefs however indicated that aid came in the form of a conflict mitigation program.**

It is possible though that these questions do not capture actual awareness. Indeed, we find that – in response to the same question - only 40 households (1.2%; 38 of the 40 living in treatment communities) and none of the chiefs confirm that the village received an infrastructure project. This stands in sharp contrast to the question that specifically related to STEP, where 60% of respondents indicated to have at least heard about the infrastructure project. It thus seems that **respondents only indicated knowing about an aid project if the question was formulated sufficiently specific.**

Households (not chiefs) were asked an additional question, that was a bit more specific, i.e. "Are you aware of the work carried out in your village / locality?", followed by four answer possibilities: 'Infrastructure construction', 'Conflict resolution activity', 'Aid distribution' and 'Other'. In response to this question, 32% of households acknowledged the infrastructure works, among which 20% in control and 39% in CDD (+). However, **only 31 households (1%) recognize the conflict mitigation project. Moreover, 29 of these households live in the control communities, and just 1 in a CDD+ community**.

Another way to verify whether the contracted NGOs implemented the intervention is by exploring the survey data on conflicts. Both households and chiefs were asked whether in the last month, a specific type of conflict occurred in the village. Seven conflict types were defined: a land conflict, a dowry conflict, an ethnic conflict, burglaries and theft of property, physical assault, domestic violence, and armed robbery & murder. When the answer was affirmative, three follow-up questions were asked: how many such conflicts, how many were resolved peacefully, and who resolved the conflict. An 'NGO' features among the coded answers to this last question.

Combined, all 3372 households of the control and CDD (+) communities mention that 7512 within-village conflicts occurred in the past month, of which 2666 (35%) were peacefully resolved, but **just 1 of the respondents mentioned that the conflict resolution was thanks to an NGO**. This respondent was an inhabitant of a CDD community, not a CDD+ community. In 723 cases (27%) the chief is mentioned as the successful conflict mediator. In the vast majority of the remaining cases, other local actors are mentioned (such as the village elders, or the chief of the locality).

This is not sufficient proof that the NGO did not implement the conflict mitigation treatment, as the idea was that the NGO would increase the conflict resolution capacity of local actors, rather than intervene directly. Nevertheless, as the chiefs are reported to be the main conflict mediators, we would have expected that at least the chiefs would have mentioned NGOs more often, but they did not. All 330 control/CDD (+) chiefs combined mentioned 1409 conflicts, among which 514 (36%) were reported as having been peacefully solved; 18% by the chiefs themselves, none by an NGO.

It thus seems fair to say that the conflict mitigation intervention did not make a big splash at the grassroots level. In fact, it is possible that – in the absence of close oversight – this intervention was not properly implemented by the NGOs contracted for this task. Indeed, while an infrastructure project leaves visible traces when (properly) executed, and is thus easier to monitor, this is not the case for a conflict mitigation intervention where no specific activities were imposed (apart from conflict mapping and report writing). That it is easier to cheat regarding activities that do not leave physical traces is also confirmed by Olken (2007) and Bertrand, et al. (2007).

Alternatively, it could be that the conflict mitigation treatment was properly implemented, but at a higher administrative level, e.g., of the *groupement* or the *territoire*. This could be the appropriate level in case of conflicts involving mobile armed groups, or in case of between, rather than within-village conflicts. However, this would have entailed spillovers from CDD+ to control communities, and thus be difficult to detect on our impact evaluation.

We now move to present and discuss quantitative results on the effects of the two interventions under study (i.e., CDD and CDD+.

5. Validity of the experiment and data collection

Before we present results, we discuss possible threats to this study's internal validity.

5.1 Balance

Randomization should ensure that control and beneficiary communities are similar. We check for balance on each of the outcome variables in the ex-ante survey, as well as on ex-post individual- and village-level variables that are unlikely to be influenced by the treatment. We do so for the primary communities, i.e. those initially assigned to the different experimental arms.

In Table 11, we show that several of the baseline characteristics are well-balanced across treatment and control, and across the two treatment arms. However, some characteristics are imbalanced across the different groups.

TABLE 11. BALANCE ON BASELINE CHARACTERISTICS

Variable (INDEX)	N [Clusters]	Control (SD)	N [Clusters]	Treatment CDD (SD)	N [Clusters]	Treatment CDD+ (SD)	Diff. Ctrl – Treat CDD (SE)	Diff. Ctrl – Treat CDD+ (SE)	Diff. Treat CDD – Treat CDD+ (SE)
Source: Household Survey									
Income Generating Activities	1065 [107]	0.00 (0.057)	1118 [113]	-0.080 (0.055)	1146 [115]	0.038 (0.069)	0.078 (0.088)	0.15 (0.096)	0.11 (0.083)
Distance to Infrastructure	1051 [106]	0.00 (2.43)	1080 [112]	0.337 (6.39)	1103 [112]	0.060 (3.021)	0.27 (0.22)	0.14 (0.14)	-0.32 (0.22)
Use of Quality Infrastructure	1034 [107]	0.00 (2.007)	1101 [113]	0.014 (2)	1116 [115]	0.012 (2.056)	-0.11 (0.10)	-0.065 (0.11)	0.13 (0.08)
Presence of Infrastructure	1062 [107]	0.00 (2.54)	1104 [113]	0.350 (3.2)	1133 [115]	0.293 (2.94)	0.056 (0.14)	0.21 (0.14)	-0.069 (0.12)

Use Infrastructure	226 [50]	0.00 (1.61)	[113] [59]	0.085 (2.73)	359 [71]	0.101 (2.3)	-0.099 (0.17)	0.074 (0.19)	-0.15 (0.19)
Need of Infrastructure	832 [98]	0.00 (2.7)	703 [92]	-0.229 (3.18)	738 [97]	-0.143 (2.93)	-0.33** (0.15)	-0.12 (0.15)	0.11 (0.16)
Quality of Infrastructure	917 [102]	0.00 (2.9)	873 [93]	-0.113 (3.14)	963 [101]	0.215 (2.19)	0.0011 (0.16)	0.37*** (0.13)	0.34*** (0.12)
Cohesion	1065 [107]	0.00 (1.88)	1118 [113]	-0.323 (2.78)	1146 [115]	-0.258 (2.6)	-0.24** (0.11)	-0.10 (0.10)	0.11 (0.11)
Inclusion	1061 [107]	0.00 (1.84)	1117 [113]	-0.232 (2.49)	1145 [115]	-0.207 (2.6)	-0.23** (0.11)	-0.18* (0.11)	0.043 (0.10)
Intra-Village Trust	1063 [107]	0.00 (2.13)	1118 [113]	-0.274 (2.66)	1144 [115]	-0.046 (2.01)	-0.082 (0.11)	0.17** (0.084)	0.24*** (0.092)
Inter-Village Trust	1054 [107]	0.00 (2.05)	1103 [113]	0.034 (2.16)	1132 [115]	0.111 (2.26)	-0.091 (0.10)	0.16 (0010)	0.12 (0.09)
Number of divisions	682 [100]	0.00 (2.18)	713 [111]	0.037 (1.96)	686 [109]	0.112 (2.35)	-0.0079 (0.13)	0.072 (0.14)	0.062 (0.11)
Number of divisions driving violence	776 [102]	0.00 (2.04)	782 [113]	0.037 (2.05)	800 [111]	0.034 (2.31)	-0.025 (0.12)	0.038 (0.13)	-0.013 (0.11)
Governance and Participation	632 [94]	0.00 (2.24)	649 [97]	0.147 (3.02)	747 [104]	-0.036 (1.73)	0.22 (0.19)	-0.022 (0.13)	-0.25 (0.16)
Source: Chief and Household Survey									
Conflict within village	1065 [107]	0.00 (2.26)	1118 [113]	0.013 (2.016)	1146 [115]	0.160 (2.702)	-0.071 (0.11)	0.087 (0.12)	0.14 (0.098)
Conflict between villages	1065 [107]	0.00 (2.23)	1118 [113]	-0.072 (1.681)	1146 [115]	0.091 (2.397)	-0.12 (0.1)	0.076 (0.12)	0.18** (0.086)

Note: Tests of difference based on OLS regression with year and province fixed effects, errors clustered at the CDD level using the household and chief surveys. The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level.

Table 12 shows that there is almost perfect balance on (quasi-)time-invariant end-line characteristics (not affected by the treatment). To control for invariant characteristics and address some imbalance, we include in our vector of covariates X_{ihv} , including the age, gender, marital status. and literacy of the respondents.

TABLE 12. BALANCE ON ENDLINE CHARACTERISTICS

Variable	N [Clusters]	Control (SD)	N [Clusters]	Treatment CDD (SD)	N [Clusters]	Treatment CDD+ (SD)	Difference Ctrl – Treat CDD (SE)	Difference Ctrl – Treat CDD+ (SE)	Difference Treat CDD – Treat CDD+ (SE)
Source: Household Survey									
Gender	1179 [118]	0.612 (0.602)	1101 [111]	0.603 (0.626)	1092 [111]	0.619 (0.595)	-0.041 (0.16)	0.023 (0.031)	0.017 (0.025)
Age	1175 [118]	43.352 (21.30)	1099 [111]	42.924 (18.819)	1090 [111]	41.784 (19.011)	-1.017 (0.95)	-1.57 (0.99)	-1.15 (0.80)
Literacy	1179 [118]	0.751 (0.667)	1100 [111]	0.757 (0.670)	1092 [111]	0.735 (0.710)	-0.0025 (0.034)	-0.0018 (0.035)	-0.023 (0.28)
Marital Status (1 if married)	1178 [118]	0.747 (0.543)	1099 [111]	0.780 (0.564)	1092 [111]	0.766 (0.558)	0.027 (0.026)	0.038 (0.028)	-0.013 (0.022)
Source: Chief Survey									
Gender	115	0.930 (0.256)	109	0.982 (0.135)	106	0.953 (0.213)	0.048 (0.036)	0.052 (0.032)	-0.028 (0.024)
Age	115	52.704 (12.85)	109	51.881 (12.294)	106	50.311 (13.270)	-0.48 (2.14)	-0.74 (2.084)	-1.75 (1.78)
Literacy	104	0.981 (0.138)	99	0.949 (0.220)	100	0.950 (0.219)	-0.014 (0.031)	-0.036 (0.037)	0.00097 (0.033)
Marital Status (1 if married)	104	0.952 (0.215)	99	0.929 (0.258)	100	0.930 (0.256)	-0.073 (0.052)	-0.07 (0.048)	0.0084 (0.036)

Note: Tests of difference based on OLS regression with year and province fixed effects, errors clustered at the CDD level using the household and chief surveys. *** (**) [*] indicates significance at the 99% (95%) [90%] level.

5.2 Attrition

We were unable to collect ex-ante and ex-post survey data for every community in our sample, either due to insecurity, inaccessibility and other operational challenges.

First, 35 projects were dropped from the initially randomized sample due to security issues or because the community implement the infrastructure before the ex-ante survey could take place. This is the case for 18 (13%) CDD communities, 13 (9.6%) CDD+ communities, and 4 (3%) control communities. The imbalance across experimental groups is mainly due to CDD(+) communities dropping out because they already started infrastructure works prior to the ex-ante survey. These dropouts are thus likely to be among the best-performing communities, biasing our results to zero rather than upward.

Second, among 27 of the 365 communities targeted in the data collection, the ex-ante data was not collected or lost, because the project was launched before the ex-ante survey, due to insecurity, or because of accidents in the field. Finally, for 25 of these 365 communities, no ex-post data is available, because of budgetary constraints, or the inaccessibility of the zone due to insecurity or poor road conditions.

Since our baseline analysis relies primarily on the ex-post data, we end up with 340 out of the 365 targeted communities in our final analytical sample. This amounts to 6.9% attrition. The attrition is higher among treatment communities than among control communities, at 7.5% and 9.0%, for CDD and CDD+ respectively, versus 4.1% for control communities. Table 13 gives an overview of attrition in the primary sample. Table 14 gives an overview of the reasons for the drop-outs, as reported by field agents.

TABLE 13. ATTRITION

Type	A. Targeted in Surveys	B. Total Survey Ex-post	C. Missing Ex-post	D. Missing Ex-ante	% Not Surveyed (C+D)/A	% Ex-post Not Surveyed (C/A)
CDD	120	111	9	7	13.3%	7.5%
CDD+	122	111	11	7	14.8%	9.0%
Control	123	118	5	13	14.6%	4.1%
TOTAL	**365**	**340**	**25**	**27**	**14.2%**	**6.9%**

In addition, the field agents reported reasons for which the communities would attrite. These reasons -non-exhaustive list- are reported in the table below

TABLE 14. REASONS FOR THE DROP-OUTS

REASONS EX-ANTE NOT DONE	
Security issue	7
Loss of data	8
Project launched before data collection	6
Abandon of project by the FSRDC	12
TOTAL	**33**
REASONS EX-POST NOT DONE	
Inaccessibility	1
Abandon of project by the FSRDC	6
Project already existing in community	11
Community not found	3
No reason	7
Security issue	5
Cost	3
TOTAL	**36**

Differential non-random attrition across treatment and control may lead to biased results. To investigate the extent and nature of attrition bias, we model attrition as a function of preprogram outcome variables, time-invariant variables as well as treatment assignment. In particular, we estimate a linear probability model that regresses an attrition indicator on the treatment indicator interacted with key ex-ante outcome variables (conflicts, social cohesion and infrastructure) and a set of strata fixed effects (year, province, and type of project fixed effects), along with control variables. We find that attritors are not significantly different from the non-attritors (except for a slightly higher educational infrastructure, and slightly lower between-village conflict incidence).

TABLE 15. DETERMINANTS OF ATTRITION

	INDEX Infrastructure	Number of Infrastructure in Village	Use of Infrastructure in Village	Distance to Infrastructure	Health Infrastructure	Educational Infrastructure
[Var]	-0.0096 (0.012)	-0.01 (0.013)	-0.012 (0.0092)	0.019 (0.025)	-0.025 (0.022)	0.027** (0.012)
CDD	0.077** (0.038)	-0.0075 (0.062)	0.079** (0.039)	0.082** (0.04)	0.087** (0.04)	0.082** (0.04)
CDD * [Var]	0.045* (0.025)	0.048* (0.027)	0.036* (0.021)	-0.017 (0.025)	-0.00038 (0.036)	-0.037 (0.031)
CDD +	0.1** (0.042)	0.12* (0.072)	0.094** (0.041)	0.09** (0.041)	0.092** (0.041)	0.093** (0.042)
CDD + * [Var]	-0.011 (0.022)	-0.0118 (0.0240)	0.015 (0.02)	-0.0036 (0.033)	0.028 (0.03)	-0.036 (0.037)
Observations	3315	3315	3315	3222	3308	3307

	INDEX Conflict Within	INDEX Conflict Between	INDEX Cohesion	INDEX Inclusion
[Var]	-0.001 (0.023)	-0.022** (0.0095)	0.0076 (0.014)	0.00019 (0.016)
CDD	0.082** (0.04)	0.084** (0.04)	0.078** (0.039)	0.082** (0.04)
CDD * [Var]	-0.018 (0.027)	0.034 (0.022)	-0.023 (0.019)	-0.0023 (0.021)
CDD+	0.093** (0.041)	0.094** (0.041)	0.088** (0.04)	0.09** (0.041)
CDD + * [Var]	-0.012 (0.027)	0.0097 (0.018)	-0.029 (0.019)	-0.012 (0.023)
Observations	3315	3315	3310	3310

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures). Based on the ex-ante sample of 3,333 households

5.3 Compliance

Our main analysis focuses on estimating ITT, thereby considering any selected CDD (+) community as a program beneficiary, regardless of whether the program was implemented or not. Conversely, any control community is considered as a non-beneficiary community. In reality, take-up rates are rarely 100%. In our case, we find take-up rates of 82% among selected CDD and CDD+ respectively: these communities put in place the infrastructure prior to the ex-post survey. In 15% of CDD and CDD+ communities there were considerably delays, to the extent the construction was still to be launched at the time of the survey; and in 3% of communities the construction was dropped due to a lack of funding. Among the control communities, we find 22 (18.6%) cases of communities who started the infrastructure work before the ex-post survey, this despite them being urged to wait for their turn in the roll-out schedule.

Because of the above non-compliance issues, the ITT estimate is likely to differ substantially from the Average Treatment Effect with perfect compliance. As the ITT may be too restrictive, we perform a LATE (Local Average Treatment Effect) estimation in a robustness check, thus estimating the average treatment effect on communities that actually received the treatment. The LATE yields valid estimates under several assumptions (exclusion restrictions, no "defers" assumption)[23]. To obtain the LATE, we use the random assignment as an instrumental variable for the treatment. We then perform a 2-stage least square analysis, where the random assignment is used as an instrument for benefiting from the treatment. In practice, we estimate the model once for each treatment arm, instrumenting take-up by the random assignment to CDD or CDD+.

We verify compliance with 9 variables that may corroborate the happening of the intervention. Table 16 shows the distribution of these variables across the treatment arms. Through these manipulation checks, we see that, compared with the control group, in the CDD (+) communities' infrastructure is more likely to be built, individuals are more likely to be satisfied with it and the households to use it. However, we see no comparative effects on the individual participations in identification, execution, contribution and administration of the infrastructure.

TABLE 16. MANIPULATION CHECKS

	INDEX	Know	Built	Satisfied	Individual Uses	Household Uses	Identification	Contribution	Execution	Administration
CDD	0.21* (0.11)	0.10* (0.058)	0.34*** (0.056)	5.04*** (0.84)	0.26 (0.22)	0.82** (0.26)	0.012 (0.037)	0.037 (0.035)	0.0093 (0.039)	-0.002 (0.023)
CDD +	0.39*** (0.12)	0.19*** (0.057)	0.38*** (0.056)	5.05*** (0.86)	0.69* (0.37)	1.24** (0.47)	0.0086 (0.035)	0.048 (0.036)	0.024 (0.041)	-0.011 (0.025)
Observations	3362	3362	3362	1834	3365	3365	3362	3362	3362	3362

[23] Van Der Windt, P. "10 Things to Know About the Local Average Treatment Effect", egap resources, last consulted on 22/04/2021.

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

A related issue is the 'John Henry effect'. The control communities knew treatment communities could start their project earlier. From the household survey this led to a feeling of being disadvantaged. We derive this from responses to the question 'NGOs, government and other organizations are present in the region and are implementing aid programs. If you compare your village with the other villages, would you say that you are at a disadvantage in relation to the programs." 51% Respondents in the control group concur with this statement regarding FSDRC programs, compared to 41% and 34% of CDD and CDD+ communities respectively.

TABLE 17. DISADVANTAGED RESPONDENTS WRT AID PROGRAMS

	Share of respondents that feels disadvantaged wrt aid programs of…				
	NGOs	National government	Provincial government	FSDRC	International donors
CDD	73.33%	73.66%	71.71%	40.65%	36.58%
CDD+	74.51%	74.51%	74.34%	33.63%	45.13%
Control	70.57%	85.43%	80%	51.14%	60.29%

This feeling of being disadvantaged may have led to compensation behaviour, whereby control communities put in efforts to make up for the disadvantage. This 'John Henry effect' may reduce the difference in outcomes between control and treatment communities.

5.4 Spillovers

Spillovers could also bias our results. Evidence from the impact evaluation of another CDD program in Eastern Congo (Humphreys et al., 2012) suggests that spillover concerns are not likely to be large since individuals in control communities (a) had limited awareness of the existence of the project in neighbouring villages and (b) had limited access to services from neighbouring villages.

This may be true for the CDD, but matters are likely to be different for the conflict mitigation component. Indeed, the conflict mitigation component had a spillover component in case it was implemented with the involvement of actors at a level higher-up in the administrative chain that the community, e.g., the territory level. For instance, in Lumera the NGO in charge of the conflict mitigation component worked with a mayi mayi group, and by doing so, generated (positive) spillover for other projects in the area, e.g., Bwegera. Such spill overs further reinforce the status of the ITT estimate as a lower bound for the average treatment effect.

6. Results on Program Impacts

In this section we present the study's main findings of the impact of the two interventions under study (e.g., CDD and CDD+) on key outcomes of interest. We use an array of several outcome indicators related two broader outcome families or domain highlighted in our theory of change: (i) infrastructure access and quality; and (ii) social cohesion. We divide each of them in more precise outcome families to capture a more detailed picture of the intervention effect (or lack thereof) on these key outcomes. Finally, in second instance, we also investigate impacts on (iii) socioeconomic economic well-being.

In the tables, the family of outcomes are pulled together in indices using the Kling, Liebman and Katz 2007 methodology. This allows us to easily interpret and compare the magnitude of our results. In Appendix 4 we unpack these index-level results into their individual measures. In case a result is significantly different from zero, and in line with the hypothesis as derived from the ToC, we highlight it in green. In case it is significant but not in line with the hypothesized result, it is highlighted in orange.

6.1 Impact on infrastructure

We begin with results on the infrastructural dimension of the ToC, summarized in Table 18. The specific (sub) outcome dimensions investigated include, conditional on the existence of the infrastructure: access and level of use by the household, satisfaction with the infrastructure, health, and education indicators, among others.

TABLE 18. PROGRAM IMPACT BY FAMILY: INFRASTRUCTURE

Infrastructure-related outcomes: Improving access to community social and economic infrastructure						
	Households					**Chief**
	Infrastructure access	**Infrastructure use**	**Health infrastructure**	**Education infrastructure**	**Satisfaction STEP**	**Infrastructure provision – chief survey**
Control	0.00	0.00	0.00	0.00	0.00	0.00
CDD (se)	-0.17 (0.13)	-0.24 (0.18)	-0.041 (0.14)	0.066 (0.14)		-0.092 (0.16)
CDD+ (se)	-0.086 (0.13)	-0.44** (0.18)	-0.18 (0.14)	0.077 (0.14)	0.036 (0.13)	-0.16 (0.14)
N	3362	1174	3360	3360	1350	299
Effect Direction	+	+	+	+	+	+

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures). The last row indicates in which direction the index should be read (as a positive or a negative outcome).

Impact on infrastructure: While the intervention aims at constructing infrastructure for the community to improve access and use, we do not observe any positive effects on these variables for the CDD or the CDD plus interventions. To the contrary, we observe negative effects on most infrastructure outcome variables, whilst mostly insignificant. The narratives presented in the previous section suggest some reasons for the lack of positive impact, such as incomplete infrastructure works; non-operational schools and health centers (because of lack of staff and equipment); lack of access to schools, health centers and improved wells (due to high service fees); and inconveniencies such as the absence of a road connection to the marketplace, or the absence of water access in a health center.

Another reason for the null result may relate to the large variety of projects, across six different sectors: education; energy; health and nutrition; road rehabilitation; storage and market facilities; and water and sanitation. Many different types of construction work took place, going from the building of a bridge to the enlargement of rehabilitation of a school. Since the projects were so diverse, access and use may not be easy to capture by a one-size-fits-all set of household survey questions, the more so because few people knew about the new infrastructure (46% reported not knowing the STEP project at all).[24]

6.2 Impact on social cohesion

For the social cohesion outcome dimension of the theory of change, we measured several indicators related to two subdimensions: (i) facilitating and improving inclusive community participation processes (e.g., cooperation and collective action between and within villages; inclusion of the most vulnerable in the process, etc.); and (ii) strengthening local conflict prevention and resolution mechanisms (e.g., existence of ethic division, disputes within or between villages, trust within and between villages, etc.).

In Table 19 we summarize the results of the social cohesion dimension.

[24] Regression analysis on individual variables, robustness checks and heterogenous analyses show the same tendency to null results, which confirms the difficulty to measure any impact on these variables, regardless of the specification.

TABLE 19. PROGRAM IMPACT BY FAMILY: SOCIAL COHESION

Social Cohesion related outcomes: Facilitating and improving inclusive community participation processes

Households								
	Community organization	**Information transmission**	**Trust**	**Cleavages**	**Cohesion**	**Conflict within village**	**Conflict between village**	**Social Inclusion**
Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CDD (se)	-0.22* (0.13)	-0.42 (0.12)	0.0026 (0.11)	0.13 (0.12)	-0.13 (0.11)	-0.12 (0.11)	0.05 (0.13)	-0.16 (0.11)
CDD+ (se)	-0.14 (0.11)	-0.052 (0.13)	0.14 (0.11)	0.27** (0.12)	0.11 (0.095)	-0.14 (0.12)	-0.0056 (0.14)	-0.024 (0.11)
N	2336	3168	3361	3362	3358	3665	3665	3359
Effect Direction	+	-	+	-	+	-	-	+

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures). The last row indicates in which direction the index should be read (as a positive or a negative outcome).

<u>Impact on Social Cohesion:</u> We observe few effects on social cohesion related outcomes. For instance, we see a decrease in the number of committees organized in the community and attended by household, in both treatment groups. Effects on individual outcome variables confirm this trend as shown in Table 20. These findings go against the hypothesis that the CDD intervention, particularly the CDD +, which had an add-on conflict mitigation component, would increase cohesion, and foster more cooperation within the village.

TABLE 20.PROGRAM IMPACT BY SUB FAMILY: COMMUNITY ORGANIZATION

	INDEX Community organization	Committee attended by HH	Frequent committee attendance by HH	Number of committees organized in village	Frequency of committee meetings organized in village
Control	0.00	2.11	18.8	1.70	1.70
CDD (se)	-0.22* (0.13)	-0.21 (0.20)	-4.83* (2.83)	-0.41*** (0.14)	-0.33** (0.13)
CDD+ (se)	-0.14 (0.11)	-0.35* (0.21)	-3.17 (2.44)	-0.29* (0.15)	-0.24* (0.14)
N	2336	3315	2336	2581	2026

Note: The index was created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

In parallel, we observe in Table 21 increase in cleavages but a decrease in the number of conflicts. Cleavages are defined as conflicts between different groups within the community (namely rich-poor, male-female, young-old, autochthones-newcomers, religious, ethnicity, elites). We asked participants to tell us "Since it is sometimes difficult for the inhabitants to work together because of the divisions that exist between them, what kinds of cleavages / fractures tend to create disagreements in your village?". In CDD+ communities, we find more cleavages between young and old, autochthones and newcomers, and ethnicities, and these conflicts were .21* more likely to generate violence. However, we observe that the number of conflicts reduced in both treatment arms (and significantly in the CDD group with a decrease of 0.803* point). In both cases, conflicts were less likely to be solved at the local level[25].

[25] By the chief of the chieftainship; the leader of the group; the chief of the locality; all the inhabitants of the village together; elders (committee of elders, community BARZA, etc.); a local notable other than the village chief (teacher, doctor, etc.); a religious authority.

TABLE 21. PROGRAM IMPACT BY SUB FAMILY: CLEAVAGES AND WITHIN CONFLICTS

	INDEX Cleavages	Presence cleavages	Cleavages drove violence	INDEX Conflict within village	Occurrence of conflicts	Solved by Chief	Solved Locally
Control	0.00	0.75	0.56	0.00	2.53	0.47	0.40
CDD (se)	0.13 (0.12)	0.13 (0.12)	0.27** (0.12)	-0.12 (0.11)	-0.8* (0.44)	0.06 (0.11)	-0.16* (0.09)
CDD+ (se)	0.27** (0.12)	0.10 (0.11)	0.21* (0.11)	-0.14 (0.12)	-0.62 (0.44)	0.17 (0.12)	-0.19** (0.09)
N	3362	3362	3362	3665	3665	1808	1808

Note: The index was created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

There are no straightforward explanations for the sharper cleavages within communities. It could however relate to a reporting bias: CDD+ interventions included a "conflict management training", which means that the community was brought together to discuss conflicts and given tools to handle them better. Such a training may have made people more aware of divisions, thus increasing their reporting. Biton and Salomon, 2006 observe something similar, when a peace-building training between Israelis and Palestinians increased the salience of identity and group membership.

6.3 Impact on socioeconomic well-being

For this outcome dimension, we focus on three sets of specific outcomes at the household level: (i) economic welfare such as consumption and assets ownership; (ii) employment/ income generating activities, productivity, and earnings; and (iii) subjective well-being and self-Perception of life conditions.

In Table 22, we observe almost no effects on socio-economic well-being. However, the heterogenous analyses capture more effects on those variables, and we describe them more in detail in the following section.

TABLE 22. PROGRAM IMPACT BY FAMILY: SOCIO-ECONOMIC WELL-BEING

Socio-Economic Well-Being related outcomes			
		Households	
	Economic Welfare	**Income Generating Activities**	**Subjective Well-Being**
Control	0.00	0.00	0.00
CDD (se)	0.11 (0.09)	0.048 (0.091)	-0.086 (0.079)
CDD+ (se)	0.014 (0.075)	-0.00028 (0.093)	-0.06 (0.079)
N	3362	3353	3362
Effect Direction	+	+	+

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures). The last row indicates in which direction the index should be read (as a positive or a negative outcome).

7. Robustness and extensions

Overall, our quantitative results herein suggest a combination of null results and weakly positive or negative results on most outcome variables for the households in the study villages. Moreover, several of the significant point estimates go against the hypotheses as derived from the ToC. For instance, looking at the detailed results on infrastructure (in Table 18), we observe that compared with the control group, the CDD+ group is 8.4% (*) less likely to have a meeting hall and the CDD group is 11.7% (*) less likely to have a bridge. In terms of social cohesion-related outcomes, CDD and CDD+ appear to have increased the number of cleavages within communities (0.27**) and decreased the number of committees organized in the communities (-0.04**). Many of the effects that run counter the ToC disappear when we perform the robustness checks (See below).

Yet, several factors can help explain this pattern of largely null results, including, among others: (i) potential heterogeneity across several meaningful dimensions; and (ii) IE design and analytic choices, which require some robustness checks. We discuss both issues below and intend to explore other issues in further iterations of the analysis and the report.

7.1 Heterogeneous impact of the interventions

As with many interventions of this kind, we expect the two CDD interventions under study to interact with a range of program-level, village-level, and household-level factors in influencing the outcomes. Since differential effects may affect cost-effectiveness and distributional impacts of the interventions, we identified several factors (or subgroups) that might interact with the program and along which we will investigate possible heterogeneous effect. We present here the results for our main outcome family expressed in indexes. Detailed results by independent variables are available on request.

Project Sector: Education

First, we perform an heterogenous analysis by isolating education and health projects which represent 57% and 22% of our sample projects respectively. These are the highest sector-level shares, and we expect therefore that results on the specific health- and education-infrastructural outcomes may be measured with more precision. Other project categories include energy (1.4%), road rehabilitation (4.3%), markets (8.8%) and WASH (6.5%). We do not consider these in a separate analysis because they include too few observations to have sufficient statistical power.

While we fail to detect a positive effect of the education related CDDs on the overall index of education infrastructure, the individual outcome variables that measure the school's roof and wall quality turn up significant (results not reported but available on request). Education projects are however associated with a significantly lower social cohesion including cleavages, conflicts/social cleavages, and inclusions along with worse information transmission between the chief and the community.

TABLE 23. HETEROGENOUS ANALYSIS BY PROJECT TYPE: EDUCATION

	EDUCATION	CDD	CDD * EDUCATION	CDD+	CDD+ * EDUCATION	N
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	0.26 (0.25)	-0.035 (0.19)	-0.24 (0.24)	-0.13 (0.18)	0.089 (0.23)	3362
INFRASTRUCTURE USE	0.64** (0.27)	-0.14 (0.25)	-0.17 (0.31)	-0.076 (0.24)	-0.54* (0.32)	1174
HEALTH INFRASTRUCTURE	-0.041 (0.25)	-0.04 (0.19)	0.013 (0.24)	-0.0055 (0.20)	-0.31 (0.25)	3360
EDUCATION INFRASTRUCTURE	0.37 (0.33)	0.2 (0.26)	-0.22 (0.27)	0.2 (0.27)	-0.19 (0.28)	3360
SATISFACTION STEP	0.54** (0.23)			0.16 (0.20)	-0.22 (0.27)	1350
SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	-0.021 (0.28)	-0.25 (0.19)	0.045 (0.22)	-0.24 (0.17)	0.16 (0.22)	2336
INFORMATION TRANSMISSION	-0.12 (0.16)	-0.45** (0.20)	0.51** (0.24)	-0.56*** (0.2)	0.82*** (0.25)	3168
TRUST	-0.0091 (0.2)	-0.16 (0.15)	0.27 (0.2)	0.13 (0.16)	0.00022 (0.21)	3361
CLEAVAGES	-0.036 (0.21)	-0.16 (0.18)	0.48** (0.21)	-0.046 (0.19)	0.5** (0.26)	3362

COHESION	-0.27* (0.16)	-0.23* (0.13)	0.17 (0.18)	0.072 (0.15)	0.04 (0.17)	3358
CONFLICT WITHIN VILLAGE	-0.072 (0.21)	-0.4** (0.16)	0.52*** (0.2)	-0.39** (0.16)	0.39* (0.2)	3362
CONFLICT BETWEEN VILLAGE	0.093 (0.22)	-0.055 (0.17)	0.19 (0.22)	-0.21 (0.17)	0.36 (0.22)	3362
INCLUSION	-0.3* (0.18)	-0.32** (0.15)	0.28 (0.19)	-0.18 (0.17)	0.24 (0.18)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	-0.18 (0.17)	0.092 (0.11)	0.036 (0.16)	0.085 (0.11)	-0.13 (0.14)	3362
INCOME GENERATING ACTIVITIES	0.32* (0.18)	0.18 (0.14)	-0.22 (0.17)	0.091 (0.15)	-0.14 (0.18)	3353
SUBJECTIVE WELL-BEING	-0.061 (0.153)	-0.025 (0.116)	-0.1 (0.145)	0.0354 (0.114)	-0.155 (0.14)	3362

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

Project Sector: Health

Results related to health projects also fail to yield the expected positive effect on health infrastructure. However, we find that health projects relate positively to social cohesion outcomes. For instance, the deviation in information transmission between the chief and by the community is lower in the treated groups. We also observe less cleavages (-0.589*) and less conflicts within village (-0.5*). Overall, we also see more satisfaction of the STEP project (0.8**). These distributional effects are puzzling. we plan to investigate them further in subsequent analyses.

TABLE 24. HETEROGENEOUS ANALYSIS BY PROJECT TYPE: HEALTH

	HEALTH	CDD	CDD * HEALTH	CDD+	CDD+ * HEALTH	N
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	-0.30 (0.19)	-0.21 (0.14)	0.21 (0.26)	-0.093 (0.15)	0.067 (0.24)	3362
INFRASTRUCTURE USE	-0.071 (0.14)	-0.22 (0.2)	-0.11 (0.41)	-0.52** (0.2)	0.28 (0.29)	1174
HEALTH INFRASTRUCTURE	0.1 (0.18)	-0.094 (0.15)	0.26 (0.25)	-0.24 (0.16)	0.27 (0.26)	3360
EDUCATION INFRASTRUCTURE	-0.22 (0.24)	0.14 (0.14)	-0.35 (0.33)	0.068 (0.14)	-0.028 (0.32)	3360
SATISFACTION STEP	-0.3 (0.20)			0.81** (0.29)	-0.19 (0.15)	1350
SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	-0.085 (0.17)	-0.21 (0.14)	-0.015 (0.26)	-0.21 (0.13)	0.26 (0.22)	2336
INFORMATION TRANSMISSION	0.49* (0.26)	-0.037 (0.12)	-0.5 (0.31)	0.15 (0.14)	-0.87*** (0.28)	3168
TRUST	0.038 (0.17)	-0.0043 (0.12)	0.012 (0.23)	0.079 (0.13)	0.25 (0.23)	3361
CLEAVAGES	0.28 (0.22)	0.21* (0.13)	-0.43* (0.26)	0.4*** (0.13)	-0.59** (0.28)	3362
COHESION	0.05 (0.14)	-0.14 (0.12)	0.029 (0.22)	0.14 (0.093)	-0.15 (0.21)	3358
CONFLICT WITHIN VILLAGE	0.47** (0.19)	0.034 (0.12)	-0.62*** (0.23)	-0.038 (0.13)	-0.5** (0.24)	3362
CONFLICT BETWEEN VILLAGE	0.25 (0.2)	0.11 (0.13)	-0.23 (0.27)	0.11 (0.15)	-0.43* (0.26)	3362
INCLUSION	0.17 (0.13)	-0.13 (0.12)	-0.12 (0.22)	0.081 (0.1)	-0.44* (0.23)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	-0.11 (0.1)	0.15 (0.1)	-0.2 (0.14)	-0.001 (0.086)	0.021 (0.14)	3362
INCOME GENERATING ACTIVITIES	-0.19 (0.18)	-0.026 (0.096)	0.35* (0.21)	-0.015 (0.1)	0.12 (0.22)	3353
SUBJECTIVE WELL-BEING	0.00071 (0.116)	-0.099 (0.086)	0.053 (0.17)	-0.019 (0.089)	-0.15 (0.16)	

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

Heterogeneity by pre-intervention wealth level

The second heterogeneity analysis that we performed relates to wealth. We hypothesize that poor, subsistence-based farmers might not have been able to partake in many activities because of high opportunity cost, or (small) user fees charged. The results on the individual outcome variables show indeed that wealthier groups are more likely to use the secondary education infrastructure (not reported, available on request). Wealthier groups are also more likely to be satisfied with the STEP infrastructure but less likely to participate in village committees. We also observe that among wealthier groups, earnings have significantly increased, compare with those in the control groups.

TABLE 25. HETEROGENOUS ANALYSIS BY WEALTH

	WEALTH	CDD	CDD * WEALTH	CDD+	CDD+ * WEALTH	N
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	0.13 (0.10)	-0.15 (0.16)	-0.037 (0.15)	-0.062 (0.14)	-0.053 (0.15)	3362
INFRASTRUCTURE USE	0.042 (0.10)	-0.26 (0.23)	0.012 (0.21)	-0.58*** (0.2)	0.26 (0.23)	1174
HEALTH INFRASTRUCTURE	0.085 (0.11)	-0.044 (0.17)	0.011 (0.15)	-0.16 (0.16)	-0.047 (0.15)	3360
EDUCATION INFRASTRUCTURE	0.18 (0.12)	0.062 (0.19)	0.014 (0.16)	0.13 (0.19)	-0.12 (0.15)	3360
SATISFACTION STEP	-0.21 (0.13)			-0.15 (0.13)	0.41** (0.18)	1350
SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	0.19** (0.083)	-0.11 (0.16)	-0.18 (0.12)	-0.082 (0.13)	-0.092 (0.11)	2336
INFORMATION TRANSMISSION	-0.024 (0.099)	-0.16 (0.16)	0.027 (0.14)	-0.045 (0.15)	-0.015 (0.13)	3168
TRUST	-0.051 (0.083)	-0.04 (0.13)	0.068 (0.12)	0.091 (0.14)	0.091 (0.12)	3361
CLEAVAGES	0.22** (0.083)	0.16 (0.13)	-0.055 (0.13)	0.34*** (0.13)	-0.16 (0.12)	3362
COHESION	0.016 (0.076)	-0.17 (0.14)	0.079 (0.13)	0.1 (0.11)	-0.0057 (0.11)	3358
CONFLICT WITHIN VILLAGE	0.43*** (0.092)	0.03 (0.11)	-0.24 *(0.14)	-0.051 (0.09)	-0.17 (0.13)	3362

CONFLICT BETWEEN VILLAGE	0.32*** (0.094)	0.14 (0.12)	-0.16 (0.16)	0.08 (0.12)	-0.15 (0.14)	3362
INCLUSION	-0.0021 (0.074)	-0.19 (0.14)	0.081 (0.14)	0.0066 (0.13)	-0.072 (0.12)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	0.5*** (0.071)	0.11 (0.094)	-0.012 (0.11)	-0.019 (0.081)	0.022 (0.097)	3362
INCOME GENERATING ACTIVITIES	0.19** (0.089)	0.078 (0.12)	-0.067 (0.12)	-0.016 (0.12)	0.023 (0.12)	3353
SUBJECTIVE WELL-BEING	-0.46*** (0.073)	-0.11 (0.092)	0.047 (0.11)	-0.063 (0.093)	0.041 (0.11)	3362

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

Heterogeneity by gender

Lastly, we perform an heterogenous analysis on gender. We explore whether men and women are differently affected by the intervention and its variations. We find few significant effects in our analysis.

TABLE 26.HETEROGENOUS ANALYSIS BY GENDER

	MALE	CDD	CDD * MALE	CDD+	CDD+ * MALE	N
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	-0.0041 (0.061)	-0.19 (0.14)	0.028 (0.096)	-0.044 (0.14)	-0.066 (0.092)	3362
INFRASTRUCTURE USE	-0.19 (0.11)	-0.28 (0.2)	0.053 (0.16)	-0.56*** (0.2)	0.2 (0.17)	1174
HEALTH INFRASTRUCTURE	-0.1 (0.066)	-0.1 (0.14)	0.099 (0.094)	-0.24 (0.15)	0.095 (0.098)	3360
EDUCATION INFRASTRUCTURE	-0.015 (0.084)	0.049 (0.16)	0.03 (0.11)	0.097 (0.17)	-0.031 (0.12)	3360
SATISFACTION STEP	-0.0085 (0.087)			-0.019 (0.16)	0.089 (0.11)	1350
SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	0.17** (0.072)	-0.28** (0.12)	0.095 (0.092)	-0.14 (0.11)	0.0063 (0.087)	2336
INFORMATION TRANSMISSION	-0.031 (0.081)	-0.15 (0.14)	0.0089 (0.099)	0.017 (0.15)	-0.11 (0.1)	3168
TRUST	-0.009 (0.07)	0.013 (0.12)	-0.027 (0.1)	0.11 (0.14)	0.049 (0.1)	3361

CLEAVAGES	-0.083 (0.074)	0.033 (0.13)	0.16* (0.09)	0.21 (0.14)	0.094 (0.1)	3362
COHESION	-0.019 (0.066)	-0.12 (0.12)	-0.027 (0.1)	0.096 (0.11)	0.013 (0.093)	3358
CONFLICT WITHIN VILLAGE	0.056 (0.065)	-0.16 (0.13)	0.076 (0.09)	-0.16 (0.13)	0.034 (0.084)	3665
CONFLICT BETWEEN VILLAGE	0.03 (0.069)	-0.02 (0.15)	0.11 (0.093)	0.00065 (0.15)	-0.0061 (0.086)	3665
INCLUSION	0.027 (0.066)	-0.17 (0.13)	0.032 (0.11)	0.033 (0.12)	-0.09 (0.1)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	-0.021 (0.06)	0.15 (0.11)	-0.071 (0.08)	-0.005 (0.089)	0.027 (0.086)	3362
INCOME GENERATING ACTIVITIES	0.011 (0.0744)	0.012 (0.1)	0.061 (0.1)	-0.057 (0.11)	0.092 (0.1)	3353
SUBJECTIVE WELL - BEING	0.036 (0.0685)	-0.095 (0.099)	0.015 (0.098)	-0.058 (0.098)	-0.00226 (0.0899)	3362

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

Heterogeneity by STEP awareness

Above we showed that awareness of STEP was relatively low among respondents: 46 did not know the project at all. It is possible that some outcomes are only manifested among those that were aware of the project, and thus more likely to have been exposed at least some of the 'inclusive and participatory'process. The table below show the results of the regression analysis.

TABLE 27.HETEROGENOUS ANALYSIS BY STEP KNOWLEDGE

	KNOW STEP	CDD	CDD * KNOW STEP	CDD+	CDD+ * KNOW STEP	N
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	-0.12 (0.13)	-0.3** (0.14)	0.262 (0.174)	-0.22 (0.15)	0.26 (0.18)	3362
INFRASTRUCTURE USE	0.041 (0.19)	-0.07 (0.24)	-0.301 (0.277)	-0.4* (0.24)	-0.077 (0.27)	1174
HEALTH INFRASTRUCTURE	-0.095 (0.13)	-0.093 (0.16)	0.107 (0.182)	-0.44** (0.2)	0.42** (0.21)	3360
EDUCATION INFRASTRUCTURE	-0.27** (0.13)	-0.019 (0.14)	0.200 (0.170)	-0.11 (0.16)	0.37** (0.19)	3360
SATISFACTION STEP	0.48 (0.36)			-0.33 (0.43)	0.4 (0.44)	1350

SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	0.26* (0.14)	-0.012 (0.17)	-0.437** (0.196)	-0.16 (0.13)	-0.076 (0.18)	2336
INFORMATION TRANSMISSION	0.13 (0.14)	-0.033 (0.15)	-0.232 (0.182)	0.13 (0.16)	-0.32* (0.19)	3168
TRUST	-0.058 (0.1)	-0.11 (0.12)	0.210 (0.144)	0.03 (0.14)	0.19 (0.16)	3361
CLEAVAGES	0.052 (0.11)	0.23 (0.15)	-0.196 (0.159)	0.28* (0.15)	-0.037 (0.16)	3362
COHESION	0.13 (0.093)	-0.18 (0.12)	0.0764 (0.134)	0.22* (0.11)	-0.21 (0.14)	3358
CONFLICT WITHIN VILLAGE	0.06 (0.13)	-0.1 (0.13)	0.00409 (0.163)	-0.045 (0.15)	-0.15 (0.17)	3362
CONFLICT BETWEEN VILLAGE	0.32** (0.13)	0.24 (0.15)	-0.385** (0.167)	0.25 (0.17)	-0.47*** (0.17)	3362
INCLUSION	0.094 (0.097)	-0.244* (0.129)	0.160 (0.146)	0.12 (0.12)	-0.24* (0.14)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	-0.066 (0.076)	0.012 (0.11)	0.191* (0.114)	-0.13 (0.092)	0.24** (0.11)	3362
INCOME GENERATING ACTIVITIES	-0.14 (0.1)	-0.03 (0.11)	0.170 (0.142)	-0.028 (0.11)	0.09 (0.14)	3353
SUBJECTIVE WELL-BEING	0.19** (0.079)	-0.11 (0.098)	0.00824 (0.119)	-0.14 (0.094)	0.063 (0.11)	

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

We observe that individuals knowing about the STEP project and in the CDD+ treatment group are more likely to score positively their use of the education and health infrastructure. In parallel, conflicts between villages seems to have reduced and information transmission between the villages and the chief to have improved. We have to interpret these results carefully though as STEP awareness is an endogenous variable and may relate to respondent characteristics that also drive the outcome variables (independently of the CDD intervention).

7.2 Sample extension and LATE analysis

In order to verify the relevance and accuracy of our results, we perform a couple of robustness tests. First, we include replacement communities in the ITT analysis. Second, we run a Local Average Treatment Effect (LATE) analysis to assess the impact of the two interventions on the communities which had completed the building of the funded projects before the ex-post data collection. To facilitate comparison with our baseline results, we provide those in the right panel of the tables.

We do not observe significant changes compared to the baseline results in the case of the ITT with replacement communities included in the analysis. However, the LATE analysis brings interesting additional results. Notably, we were surprised to find no effect on the presence of infrastructure, the nature of the intervention being to build such infrastructure. **In the LATE analysis, we observe that the quality of the roofs and walls is significantly better for the CDD and CDD+ groups compared with the control group. It suggests that a lot of the work performed has been on rehabilitating work to existing infrastructure.** The identification data which describes in detail the work performed does corroborate that many infrastructures already present were renovated, or few elements were added (fence, windows, classrooms, etc.). Other outcome variables remain unchanged in the LATE approach compared with the ITT approach.

TABLE 28. ROBUSTNESS CHECK: ADDING REPLACEMENT COMMUNITIES

	ITT WITH REPLACEMENT			*ITT WITHOUT REPLACEMENT*		
	CDD Effect	**CDD + effect**	**N**	**CDD Effect**	**CDD + effect**	**N**
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	-0.12 (0.11)	-0.095 (0.11)	4413	-0.17 (0.13)	-0.086 (0.13)	3362
INFRASTRUCTURE USE	-0.16 (0.14)	-0.28** (0.14)	1585	-0.24 (0.18)	-0.44** (0.18)	1174
HEALTH INFRASTRUCTURE	-0.085 (0.11)	-0.28** (0.12)	4411	-0.041 (0.14)	-0.18 (0.14)	3360
EDUCATION INFRASTRUCTURE	0.11 (0.11)	0.011 (0.19)	4410	0.066 (0.14)	0.077 (0.14)	3360
SATISFACTION STEP		0.047 (0.12)	1782		0.036 (0.13)	1350
SOCIAL COHESION RELATED OUTCOMES						
INFRASTRUCTURE PROVISION – CHIEF SURVEY	3.31 (7.86)	-8.86 (7.12)	398	-0.092 (0.16)	-0.16 (0.14)	299
COMMUNITY ORGANIZATION	-0.12 (0.11)	-0.092 (0.1)	3070	-0.22* (0.13)	-0.14 (0.11)	2336
INFORMATION TRANSMISSION	-0.026 (0.11)	0.024 (0.11)	4196	-0.42 (0.12)	-0.052 (0.13)	3168
TRUST	0.034 (0.089)	0.14 (0.093)	4412	-0.0026 (0.12)	0.14 (0.11)	3361

CLEAVAGES	0.075 (0.099)	0.072 (0.099)	4413	0.13 (0.12)	0.27** (0.12)	3362
COHESION	-0.08 (0.087)	0.092 (0.088)	4406	-0.13 (0.11)	0.11 (0.095)	3358
CONFLICT WITHIN VILLAGE	-0.034 (0.099)	-0.088 (0.098)	4811	-0.12 (0.11)	-0.14 (0.12)	3665
CONFLICT BETWEEN VILLAGE	0.012 (0.11)	-0.0083 (0.11)	4811	0.05 (0.13)	-0.0056 (0.14)	3665
INCLUSION	-0.16* (0.09)	-0.072 (0.097)	4407	-0.16 (0.11)	-0.024 (0.11)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	0.012 (0.071)	-0.075 (0.067)	4409	0.11 (0.09)	0.014 (0.075)	3362
INCOME GENERATING ACTIVITIES	0.11 (0.077)	0.049 (0.08)	4403	0.048 (0.091)	-0.00028 (0.093)	3353
SUBJECTIVE WELL-BEING	-0.039 (0.067)	-0.031 (0.068)	4413	-0.086 (0.079)	-0.06 (0.079)	3362

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures). The sample includes all the primary targeted communities plus the replacement communities.

Table 29. Robustness check: Local average treatment effect (LATE)

	LATE (WITH REPLACEMENT)			***ITT WITHOUT REPLACEMENT***		
	CDD Effect	**CDD + effect**	**N**	**CDD Effect**	**CDD + effect**	**N**
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	-0.19 (0.12)	0.034 (0.12)	4413	-0.17 (0.13)	-0.086 (0.13)	3362
INFRASTRUCTURE USE	-0.13 (0.14)	-0.34** (0.17)	1585	-0.24 (0.18)	-0.44** (0.18)	1174
HEALTH INFRASTRUCTURE	0.035 (0.14)	-0.0077 (0.14)	4411	-0.041 (0.14)	-0.18 (0.14)	3360
EDUCATION INFRASTRUCTURE	0.077 (0.16)	0.13 (0.13)	4410	0.066 (0.14)	0.077 (0.14)	3360
SATISFACTION STEP		0.036 (0.11)	1395		0.036 (0.13)	1350
INFRASTRUCTURE PROVISION – CHIEF SURVEY	-0.07 (0.16)	-0.11 (0.15)	393			
SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	0.0075 (0.11)	0.11 (0.1)	3070	-0.22* (0.13)	-0.14 (0.11)	2336

INFORMATION TRANSMISSION	-0.025 (0.12)	0.078 (0.13)	4196	-0.42 (0.12)	-0.052 (0.13)	3168
TRUST	0.088 (0.1)	0.15 (0.11)	4412	-0.0026 (0.12)	0.14 (0.11)	3361
CLEAVAGES	0.11 (0.11)	0.3** (0.12)	4413	0.13 (0.12)	0.27** (0.12)	3362
COHESION	0.016 (0.11)	0.061 (0.11)	4406	-0.13 (0.11)	0.11 (0.095)	3358
CONFLICT WITHIN VILLAGE	-0.051 (0.1)	-0.063 (0.12)	4811	-0.12 (0.11)	-0.14 (0.12)	3665
CONFLICT BETWEEN VILLAGE	-0.035 (0.11)	-0.04 (0.14)	4811	0.05 (0.13)	-0.0056 (0.14)	3665
INCLUSION	0.13 (0.11)	0.036 (0.12)	4407	-0.16 (0.11)	-0.024 (0.11)	3359
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	0.278*** (0.077)	0.15** (0.072)	4409	0.11 (0.09)	0.014 (0.075)	3362
INCOME GENERATING ACTIVITIES	-0.036 (0.088)	-0.13 (0.095)	4403	0.048 (0.091)	-0.00028 (0.093)	3353
SUBJECTIVE WELL-BEING	0.048 (0.075)	0.025 (0.081)	4413	-0.086 (0.079)	-0.06 (0.079)	3362

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures). The CDD and CDD+ groups include (1) the primary targeted communities which completed the project, (2) the completed replacement communities. In addition, the CDD group includes (3) the controls which did complete the project. The control group includes all the communities that did not complete the project at the time of data collection (implementation status being “to be launched”, “ongoing”, “dropped”).

7.3 Difference-in-Difference analysis

We implement a difference in difference analysis to evaluate the effect of the CDD intervention on our outcomes of interest over time. The identifying assumption is that, in the absence of the CDD program, the outcome variables would have evolved in the same way across treatment and control communities.

TABLE 30. DIFFERENCE IN DIFFERENCE ANALYSIS

	EXPOST	CDD	CDD * EXPOST	CDD+	CDD+ * EXPOST	N
INFRASTRUCTURE RELATED OUTCOMES						
INFRASTRUCTURE ACCESS	0.49*** (0.11)	0.2 (0.12)	-0.52*** (0.15)	0.23* (0.13)	-0.42*** (0.15)	6020
INFRASTRUCTURE USE	-0.14 (0.19)	-0.19 (0.22)	0.093 (0.28)	-0.26 (0.21)	0.016 (0.26)	1935
HEALTH INFRASTRUCTURE	0.22** (0.095)	0.16 (0.14)	-0.23* (0.13)	0.11 (0.13)	-0.3** (0.13)	6011
EDUCATION INFRASTRUCTURE	0.2** (0.097)	0.18 (0.14)	-0.15 (0.13)	0.27** (0.13)	-0.22* (0.12)	6010
SATISFACTION STEP	1.41*** (0.15)			-0.26** (0.12)	0.39* (0.21)	1978
SOCIAL COHESION RELATED OUTCOMES						
COMMUNITY ORGANIZATION	0.07 (0.13)	-0.072 (0.12)	-0.097 (0.16)	-0.075 (0.12)	-0.025 (0.16)	3947
TRUST	-0.026 (0.098)	0.0014 (0.19)	-0.021 (0.14)	0.068 (0.1)	0.038 (0.14)	6019
CLEAVAGES	-0.054 (0.089)	-0.0035 (0.11)	0.093 (0.14)	0.11 (0.12)	0.098 (0.15)	6020
COHESION	-0.12 (0.077)	-0.32*** (0.11)	0.3** (0.14)	-0.17 (0.11)	0.38*** (0.13)	6011
CONFLICT WITHIN VILLAGE		0.12 (0.23)		0.35 (0.23)		6203
CONFLICT BETWEEN VILLAGE		-0.047 (0.26)		0.13 (0.25)		6191
INCLUSION	0.025 (0.078)	-0.25** (0.1)	0.19 (0.13)	-0.15 (0.10)	0.15 (0.13)	6012
SOCIO-ECONOMIC WELL-BEING RELATED OUTCOMES						
ECONOMIC WELFARE	0.19*** (0.046)	0.11 (0.084)	-0.079 (0.075)	0.077 (0.083)	-0.14** (0.061)	6012
INCOME GENERATING ACTIVITIES	0.12 (0.091)	0.09 (0.097)	-0.045 (0.12)	0.1 (0.1)	-0.13 (0.12)	6006
SUBJECTIVE WELL-BEING	0.11 (0.079)	0.0086 (0.084)	-0.066 (0.11)	0.061 (0.083)	-0.11 (0.11)	6020

Note: The indexes were created following the Kling et al. methodology which is a standardized protocol. The control mean is thus around 0 for each measure. *** (**) [*] indicates significance at the 99% (95%) [90%] level. Standard errors clustered at the community level (individual level measures).

The DiD estimates are very similar to our baseline results of the ITT analysis. We find either null- or negative effects on infrastructure use and access. We find a significant and positive effect on social cohesion, which is somewhat different from the baseline results.

8. Conclusions and policy implications

The STEP CDD program in Eastern DRC sought to improve infrastructure access and use, and strengthen social cohesion. It was randomly assigned to about two thirds of 400 communities with eligible project proposals. The selected communities received a budget of up to $100,000 to finance an infrastructure project, as well as training to manage the project in an inclusive and participatory way. A random half of CDD communities also received a conflict mitigation component, which consisted of conflict prevention and management activities, identified and led by NGOs specialized in the matter. According to the theory of change the CDD program would lead not only to improvements in community infrastructure but also to more social cohesion. The conflict mitigation component would enhance both of these effects. **Contrary to this theory of change, our impact evaluation did not detect any significant improvements in either the infrastructure or social cohesion dimensions.**

Our null results are not entirely inconsistent with findings from other studies. By now, the impact evaluation literature on CDDs has grown to a respectable size, and abounds with null results, especially with respect to the social engineering aspect or the 'software' as Casey (2018) calls it in her review. For instance, Casey et al. (2012), in their impact evaluation of a randomized CDD program in Sierra Leone find no sustained impact on local governance quality, decision-making processes, inclusiveness or collective action. Likewise, Barron et al. (2009), looking at Aceh's CDD program (specifically targeted at conflict victims) find no convincing evidence for improvements in social cohesion or in relations between citizens and government. The same goes for a large CDD impact evaluation in eastern Congo by Humphreys et al. (2019). Null-results on the hardware (the infrastructural dimension) are less common but not unique. A case in point is the above mentioned CDD impact evaluation in eastern Congo, for which Humphreys et al., (2012) neither find better access to infrastructure and services in CDD communities. **The existing literature invokes several reasons for the null-results.** Regarding social cohesion, the reason most often mentioned is that institutional engineering from the outside hardly ever succeeds in yielding lasting behavioral changes. The most common explanations for the null on the infrastructure dimension include the small grant amounts per person, as well as the lack of infrastructure maintenance and of complementary inputs.

Based on an exploratory analysis of the household narratives and the field observations, we conjecture that our null-results on the software are most likely due to a combination of a light intervention targeted at sticky institutions, the weak capacity and oversight of the implementing partners, and the null-result on the hardware. The choice to work with existing LDCs was made to avoid bypassing actual centers of power, which were deemed necessary to take on board in order to set in motion a sustainable transition as opposed to a short-term cosmetic change in decision-making. However, this 'Trojan horse' strategy may have backfired, as power may have remained too concentrated, giving way to a business-as-usual approach in CDD programming. In addition, the existing LDCs received only limited training and assistance. The STEP budget was almost entirely allocated to the actual construction work (>75%), whereas other CDD project allocated almost half to the institutional treatment (Mansuri and Rao,2013). Moreover, in several cases, the training and follow-up were of doubtful quality, reflecting both the poor capacity of some of the local NGOs assigned with this task and weak follow-up by the FSDRC. Thus, while working with local implementing partners has its potential merits, it may result in poor implementation when partners lack the necessary capacity or incentives.

The null-results on the hardware could be driven by the failure to identify the best possible project for the largest number of people, the poor capacity of some of the local NGOs and contractors involved, and the lack of complementary inputs. Given the relatively light social engineering intervention in the decision process, there may have been a democratic deficit in the choice of the project. And, even if the project was well chosen, the poor follow-up by some NGOs and poor execution by some of the local contractors resulted in delayed as well as flawed construction works. Finally - and a recurrent concern in CDD impact evaluations – many schools and clinics built and/or rehabilitated were not effectively operational due to the lack of complementary inputs (paid teachers, medical personal, equipment, etc.), nor accessible to all due to (high) user fees.

Other potential reasons for the null result on the infrastructural dimension include the relatively small budget spent per person, the lack of precise measurement, and the lack of full compliance. While the envelop of $100,000 is rather generous compared to other CDD projects, it is relatively small when considering the size of the targeted communities, resulting in $5.5 per person. [26] Compared to measuring changes in social cohesion, measuring infrastructure outcomes seems rather straightforward. However, where other studies have relied on specific facility audits to assess the quality of the construction and services in detail (e.g. Laudati et al. (2018), Mvukiyehe and van der Windt (2020)), the measures used in this study are rather crude.

Adding to this the heterogeneity of the chosen infrastructure projects across communities (building/rehabilitating schools, classrooms, health posts, roads, market, latrines, etc.), we may lack sufficiently precise measurement tools to pick up improvements in infrastructure

[26] The maximum project cost is $100,000, of which communities need to contribute 10%. The average community size is 16,250 (6,500 adults; and 9,750 below 18 years - assuming 60% youngsters), and $90,000 divided by 6500 is $5.5. While $5.5 per person is similar to the grant per person received in the Tuungane project ($2 per person over two years at the village level, $4 per person over two years at the village cluster level; see Humphreys et al., 2012 (Humphreys, De La Sierra and Van der Windt, Social and Economic Impacts of Tuungane Final Report on the Effects of a Community Driven Reconstruction Program in Eastern Democratic REPUBLIC of Congo. 2012)) it is smaller than the $20 per person spent in Aceh, Indonesia Barron et al. 2009 (Barron, et al. 2009) and much smaller than what is spent in the millennium development villages: $120 per person per year (Mitchell et al., 2018 (Mitchell, et al. 2018, Mitchell, et al. 2018)).

presence, access and use. This is underscored by our detection of a significant effect on construction quality (of schools) in a more homogeneous subsample, i.e., communities that chose projects in the education sector – the most common category among the chosen infrastructure projects. Finally, the less than perfect compliance in treatment and control (both at about 80%) biases our results downward, as can be derived from a comparison of our Intent-To-Treat (ITT) and LATE (Local Average Treatment Effect).

The null-result on the conflict mitigation treatment may be due to the prior existence of effective conflict mitigation mechanisms for village-level conflicts (also in control communities), the poor (or total lack of) its implementation, the ineffectiveness of the treatment for conflicts with a broader geographic scope, or the inability of the impact evaluation to pick up any positive impact on the latter type of conflicts (due to treatment spillovers). The underlying assumption of the conflict mitigation treatment is that communities do not have effective mechanisms for solving conflicts. This may be true for conflicts that transcend the village level, but it is unlikely to be true for within-village conflicts, for which villagers turn to the chief as the legitimate conflict mediator. The chief may however not be well placed to solve between-village conflicts (because of a conflict of interest) or regional conflicts (because the (f)actors involved are beyond his/her reach).

Were the contracted local NGOs able to mediate or put in place effective institutions for between-village conflict resolution? It is difficult to verify this because we do not have concrete information on the actions taken by these NGOs, which were given *carte blanche* in their concrete intervention design. Somewhat disconcertedly, we find that only two households and none of the chiefs acknowledged that the village received a conflict mitigation program. Besides, even if an effective conflict mitigation intervention would have taken place at the higher administrative level (of the *territoire*), our research design may have not been able to pick it up due to positive spillover effects.

Most CDD impact studies warn against too readily generalizing their results. We are equally prudent. To start with, the challenge set out for a CDD project is daunting: to persistently alter a community's institutional DNA by a one-off outside intervention. Even aid optimists would agree that this is not a simple matter, let alone that it could be achieved by a one-size-fits-all design. Despite the daunting challenge, it is worth trying, given the firmly demonstrated importance of the institutional environment (see e.g. Acemoglu and Robinson, 2012). Given the absence of a one-size-fit-all outside solution in this department, to allow for flexible designs and sufficient local content so that context-specific bottom-up institutional innovations can arise. This concern is what drove the STEP CDD design choices: to give a lot of leeway to local actors familiar with the context, and work with existing institutions.

Our null-results suggest that this approach may have been a bridge too far in a fragile and conflict-affected region with a huge institutional capacity deficit. It does not mean however that this approach could not work in a better institutional environment, or when local implementing partners receive sufficient support (and oversight) of external actors**. Another external validity issue relates to the fact that participating communities are drawn from the population of communities that handed in a high-quality project proposal**. These are unlikely to be the highest-capacity communities with the most conflict and internal division. In this respect, our results provide us with an upper rather than a lower bound for Eastern DR Congo.

Since our diagnosis of the null-results mainly point to issues with implementation, this is also where we see scope for policy recommendations: to make sure to work with trustworthy and capable partners. Because of local content issues and scope for scaling up, these are preferably in-country (local) partners. However, especially in fragile and conflict-affected environments, these partners may need training and oversight in order to meet the implementation standards. In addition, and as derived from the narratives of the project beneficiaries, infrastructure access and use can only significantly and inclusively be improved if complementary inputs are present (staff, equipment, maintenance), and if user fees are not prohibitively high. This should be taken up as a point of attention when screening the eligibility of project. Finally, the project could be designed even more pro-poor if labor was to be recruited locally (in a cash-for-work program) and the community contribution would be revised downward from 10% to 5%.

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Appendix

Appendix 1 The sample

Due to various reasons, ex-ante or ex-post data is unavailable for some communities. Our analytical sample for the baseline analysis eventually includes 340 communities, distributed across CONTROL, CDD or CDD+ as shown in Table A 1.

TABLE A 1. VILLAGES PER TREATMENT ARM IN THE ANALYTICAL SAMPLE

Control communities	CDD Treatment Communities	
	No conflict mitigation	Conflict mitigation
A: 118 communities	B: 111 communities	C: 111 communities

Note: The Table presents the number of communities assigned to control and each of the two treatment arms, over the period 2016 to 2019

In addition to the primary list of 400 randomized CDD and CDD+ communities, 56 more CDD and 52 more CDD+ projects were randomly selected, among the list of submitted projects. These serve as replacement projects, to be used in case a primary community dropped. Following a drop-out, a replacement project would be picked in the same geographic area, and in order of the random rank allocated to it.

Including the replacement communities, the dataset includes in total 446 communities and 4,860 individuals

TABLE A 2. DESCRIPTION OF THE SAMPLE

	HOUSEHOLDS	COMMUNITIES
CONTROL	1294	118
CDD	1210	111
CDD+	1198	111
CDD REPLACEMENT	616	56
CDD + REPLACEMENT	542	52
TOTAL	4860	446

Since we are performing an Intent-To-Treat analysis, we are focusing on the sample of projects that were initially targeted, namely the controls, the CDDs and the CDDs+, regardless of the implementation status. The final sample thus includes 340 CDDs and 3702 households. We provide a description of some key characteristics in Table A 3.

TABLE A 3. CHARACTERISTICS OF THE SAMPLE RESPONDENTS

	OBS	MEAN	SD	MIN	MAX
GENDER	3,702	.64	.48	0	1
AGE	3,694	43.50	14.8	18	95
LITERACY	3,674	.77	.42	0	1
POVERTY	3,372	.49	.5	0	1
MARRIED	3,369	.76	.42	0	1
OWNER	3,369	.87	.34	0	1

Appendix 2. Variable Definitions and Summary Information

TABLE A 4.VARIABLE DEFINITIONS AND SUMMARY INFORMATION FOR OUTCOME VARIABLES

#	Family	Variable	Description	Obs.	Mean	Sd.	Min.	Max.
1	Infrastructure access	Presence of well	Binary. There is a well in the village.	3362	0.35	0.48	0	1
2	Infrastructure access	Presence of bridge	Binary. There is a bridge in the village.	3358	0.30	0.46	0	1
3	Infrastructure access	Presence of pipes	Binary. There are pipes in the village.	3357	0.38	0.48	0	1
4	Infrastructure access	Presence of (meeting) hall	Binary. There is a (meeting) hall in the village.	3344	0.13	0.33	0	1
5	Infrastructure access	Presence of church	Binary. There is a church in the village.	3369	0.94	0.24	0	1
6	Infrastructure access	Infrastructures with high-quality wall	Continuous (0-2). Number of infrastructures with high quality wall: (meeting) hall, church. Options: 1) Earth / mud, 2) Plastic cover, 3) Clay bricks (not baked), 4) Wood / bamboo, 5) Stones, 6) Semi-durable (Pebbles, sand, cement and tree sticks), 7) Baked bricks, 8) Cement / concrete, 9) Metal sheets, 10) Cardboard. High quality are: 7 and 8.	3372	0.61	0.62	0	2
7	Infrastructure access	Infrastructures with high-quality roof	Continuous (0-2). Number of infrastructures with high quality roof: (meeting) hall, church. Options: 1) Earth, 2) Leaves, 3) Straw, 4) Wood, 5) Metal sheets, 6) Cement / concrete, 7) Tiles, 8) Plastic cover. High quality are: 5, 6, 7.	3372	0.92	0.53	0	2
8	Infrastructure use	Frequency of use of infrastructures	Continuous. How many times/days per month does the HH use following infrastructures: well, bridge, pipes, (meeting) hall, church. Sum of frequency across 5 infrastructure types.	3372	34.48	32.66	0	224
9	Health infrastructure	Presence of health centre	Binary. There is a health centre in the village.	3369	0.77	0.42	0	1

10	Health infrastructure	Health center with high-quality wall	Binary. Health centre high quality wall: (meeting) hall, church. Options: 1) Earth / mud, 2) Plastic cover, 3) Clay bricks (not baked), 4) Wood / bamboo, 5) Stones, 6) Semi-durable (Pebbles, sand, cement and tree sticks), 7) Baked bricks, 8) Cement / concrete, 9) Metal sheets, 10) Cardboard. High quality are: 7 and 8.	2483	0.64	0.48	0	1
11	Health infrastructure	Health center with high-quality roof	Binary. Health centre has high quality roof: (meeting) hall, church. Options: 1) Earth, 2) Leaves, 3) Straw, 4) Wood, 5) Metal sheets, 6) Cement / concrete, 7) Tiles, 8) Plastic cover. High quality are: 5, 6, 7.	2485	0.93	0.25	0	1
12	Health infrastructure	Frequency of use of health infrastructure	Continuous. How many times/days per month does the HH use the health centre.	2384	2.51	4.13	0	30
13	Health infrastructure	Walking time to nearest health post	Continuous. Time in minutes to walk to nearest health post during rainy season	3322	30.79	62.73	0	1830
14	Health infrastructure	Walking time to nearest health centre	Continuous. Time in minutes to walk to nearest health centre during rainy season.	3364	36.65	92.06	0	2700
15	Health infrastructure	Walking time to nearest hospital	Continuous. Time in minutes to walk to nearest hospital during rainy season.	3342	151.15	428.14	0	12030
16	Health infrastructure	Seeks health care in health care facility	Binary. Where does HH usually seeks health care options : (1) State health post or clinic or hospital (public); (2) Health post or clinic or hospital of a church or an NGO; (3) Health post or clinic or private hospital; (4) Pharmacy; (5) Traditional healer. Options 1, 2 and 3 equal one.	3337	0.77	0.42	0	1
17	Education infrastructure	Presence of primary school	Binary. There is a primary school in the village.	3369	0.91	0.28	0	1
18	Education infrastructure	Presence of secondary school	Binary. There is a secondary school in the village.	3368	0.85	0.36	0	1
19	Education infrastructure	Primary school with high-quality wall	Binary. Primary school has high quality wall: (meeting) hall, church. Options: 1) Earth / mud, 2) Plastic cover, 3) Clay bricks (not baked), 4) Wood / bamboo, 5) Stones, 6) Semi-durable (Pebbles, sand, cement and tree sticks), 7) Baked bricks, 8) Cement / concrete, 9) Metal sheets, 10) Cardboard. High quality are: 7 and 8.	2947	0.59	0.49	0	1
20	Education infrastructure	Primary school with high-quality roof	Binary. Primary school has high quality roof: (meeting) hall, church. Options: 1) Earth, 2) Leaves, 3) Straw, 4) Wood, 5) Metal sheets, 6) Cement / concrete, 7) Tiles, 8) Plastic cover. High quality are: 5, 6, 7.	2959	0.90	0.31	0	1
21	Education infrastructure	Secondary school with high-quality wall	Binary. Secondary school has high quality wall: (meeting) hall, church. Options: 1) Earth / mud, 2) Plastic cover, 3) Clay bricks (not baked), 4) Wood / bamboo, 5) Stones, 6) Semi-	2719	0.57	0.50	0	1

			durable (Pebbles, sand, cement, and tree sticks), 7) Baked bricks, 8) Cement / concrete, 9) Metal sheets, 10) Cardboard. High quality are: 7 and 8.					
22	Education infrastructure	Secondary school with high-quality roof	Binary. Secondary school has high quality roof: (meeting) hall, church. Options: 1) Earth, 2) Leaves, 3) Straw, 4) Wood, 5) Metal sheets, 6) Cement / concrete, 7) Tiles, 8) Plastic cover. High quality are: 5, 6, 7.	2731	0.89	0.31	0	1
23	Education infrastructure	Walking time to nearest primary school	Continuous. Time in minutes to walk to nearest primary school.	3365	22.61	96.97	0	3050
24	Education infrastructure	Walking time to nearest secondary school	Continuous. Time in minutes to walk to nearest secondary school.	3365	28.47	99.39	0	2745
25	Education infrastructure	School-aged children go to (high-quality) school	Continuous. Which type of school children of the HH are attending. (1) Official school (public), (2) Private non-religious school, (3) Catholic convention school, (4) Protestant convention school, (5) School organized by an NGO, (6) None (don't go to school). Options (1) and (3) equal 2 (high quality), Options (2) and (4) equal 2 (low quality), option (6) equals 0.	1992	1.60	0.63	0	2
26	Satisfaction STEP	Satisfaction with the CDD process	Continuous (0-18). Summation across 18 different categories: 4 positively formulated: Are you satisfied with the initiative taken to build this structure? With the quality of the work? With the Building? With the collaboration between communities that it entailed? (0) totally disagree, (1) disagree, (2) agree, (3) totally agree. Options 2 and 3 equal 1. 14 negatively formulated: Q106 The whole process (project decision and execution) took too long Q107 The ALE (NGO in charge of the project) did not behave well in the village Q108 The selected projects were not the most important for this village Q109 The selected projects did not help enough people in this village Q110 I had no real influence on the selection of projects Q111 Disagreements about the project in this village were not well managed Q112 The process required too many steps / formalities to follow Q113 There was not enough information about the process	1839	8.50	6.10	0	18

			Q114 There has been embezzlement in this village (corruption, nepotism, etc.) Q115 The allocation of money across villages was not done fairly Q116 The project created divisions or conflicts in the community Q117 The project created divisions or conflicts between the communities Q118 The project was controlled by the manager Q119 The project did not reflect your concerns (0) totally disagree, (1) disagree, (2) agree, (4) totally agree. Options 0 and 1 equal 1.					
27	Satisfaction STEP	Personal use of STEP project	Continuous. Last week, how many times did you personally use the STEP infrastructure?	1368	1.64	4.02	0	30
28	Satisfaction STEP	Household use of STEP project	Continuous. Last week, how many times did you personally or your household members use the STEP infrastructure?	1391	3.15	5.70	0	50
29	Infrastructure provision – chief survey	Number of wells	Continuous. Number of wells in the chief's entity.	326	3.00	4.66	0	29
30	Infrastructure provision – chief survey	Number of (meeting) halls	Continuous. Number of (meeting) halls in the chief's entity.	319	0.25	0.62	0	3
31	Infrastructure provision – chief survey	Number of churches	Continuous. Number of churches in the chief's entity.	324	3.88	3.20	0	16
32	Infrastructure provision – chief survey	Number of health centers	Continuous. Number of health centers in the chief's entity.	326	1.40	1.34	0	9
33	Infrastructure provision – chief survey	Number of schools (classes)	Continuous. Number of classes (in school) in the chief's entity.	317	23.61	23.74	0	120
34	Infrastructure provision – chief survey	Number of infrastructure obtained on request of village	Continuous. Number of wells / (meeting) halls / churches / health centres / schools (classes) in the chief's entity, obtained at initiative of entity.	330	17.85	22.62	0	134
35	Infrastructure provision – chief survey	Number of infrastructure obtained at initiative of state of NGO without village contribution	Continuous. Number of wells / (meeting) halls / churches / health centres / schools (classes) in the chief's entity obtained at the initiative of the State or an NGO without the material or financial support of your entity?	330	8.36	14.78	0	84
36	Infrastructure provision – chief survey	Number of infrastructure obtained at initiative of state of NGO with village contribution	Continuous. Number of wells/ (meeting) halls / churches / health centres / schools (classes) in the chief's entity obtained at the initiative of the State or an NGO with the material or financial support of your entity?	330	4.69	8.05	0	53
37	Community organization	Committee attended by HH	Continuous (0-8). Household participated in health committee, education committee, agriculture committee, security committee,	2585	1.49	1.29	0	8

			conflict committee, governance committee, human rights committee, other governance committee. Sum of types of committees in which household participated.					
38	Community organization	Frequent committee attendance by HH	Continuous (0-8). Frequency of household participated in health committee, education committee, agriculture committee, security committee, conflict committee, governance committee, human rights committee, other governance committee. Sum of types of committees in which household participated. (0) rarely, (1) from time to time (2) always. Options 1 and 2 equal 1.	2028	1.53	1.14	0	8
39	Community organization	Number of committees organized in village	Continuous (0-8). Village has health committee, education committee, agriculture committee, security committee, conflict committee, governance committee, human rights committee, other governance committee. Sum of different types of committees that exist in village.	3323	2.02	1.67	0	8
40	Community organization	Frequency of committee meetings organized in village	Continuous. Number of meetings of education committee, agriculture committee, security committee, conflict committee, governance committee, human rights committee, other governance committee. Sum of meetings across different types of committees.	2339	17.72	21.66	0	264
41	Information transmission	Deviation between chief's and HH answers wrt aid received	Received financial or food or social assistance aid from the government, NGOs or donors.	3177	0.11	0.32	0	1
42	Information transmission	Deviation between chief's and HH answers wrt source of aid received	Name of the aid program	3372	0.05	0.25	0	3
43	Information transmission	Deviation between chief's and HH answers wrt amount of the aid received	Amount of the aid	3372	51522 53	39000 000	0	74000 0000
44	Information transmission	Deviation between chief's and HH answers wrt perception of favoritism to other villages	Perception of favoritism in aid allocation compare to other villages	1503	0.96	1.23	0	5
45	Trust	Intra-group trust	Continuous. If you want to shop in the village or market but are unable to make it, is there ... [a member of	3370	2.03	0.84	0	3

			the following groups] ready to do it for you? Q60 a member of my family Q61 a member of this village Q63 someone who is not from my village but who is from my ethnic group (0)no, (1) maybe, (2) probably, (3) certainly. 2 and 3 equal 1, summed across 3 categories.					
46	Trust	Inter-group trust	Continuous. If you want to shop in the village or market but are unable to make it, is there ... [a member of the following groups] ready to do it for you? Q62 a member from another village Q64 someone who is neither from my village nor from my ethnic group Q65 an ex-combatant (0)no, (1) maybe, (2) probably, (3) certainly. 2 and 3 equal 1, summed across 3 categories.	3370	0.81	1.00	0	3
47	Cleavages	Within-village cleavages	Continuous (0-7) Q66: What are the cleavages/divergence that creates disagreement in your village? Sum over all the 7 items	3372	0.79	1.07	0	7
48	Cleavages	Cleavages driving violent conflicts	Continuous (0-7) Q68 : Did these divisions drive violent events in the last 6 months. Binary. 0 No 1 Yes. Replace missing by 0 as to say “Where there conflicts with violence”. Items: q66_divisiontype_poor_rich: between the richest and the poorest? q66_divisiontype_men_women: between men and women? q66_divisiontype_young_old: Between young and old? q66_divisiontype_autoch_new: Between natives and newcomers? q66_divisiontype_religion: Between religions? q66_divisiontype_ethnic: Between tribes or ethnic groups? q66_divisiontype_elites: Between different elites?	3372	0.56	0.93	0	6
49	Cohesion	Attitudes towards returned internally displaced	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that returned internally displaced Live in village; Acquire land title; Become settlement chief; Become close friend; Marry a family member.	3367	4.43	0.88	0	5
50	Cohesion	Attitudes towards returned refugees	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that returned refugees Live in village; Acquire land title; Become	3363	4.02	1.28	0	5

			settlement chief; Become close friend; Marry a family member.					
51	Cohesion	Attitudes towards individuals of other ethnic groups	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that individual belonging to other ethnic group Live in village; Acquire land title; Become settlement chief; Become close friend; Marry a family member.	3360	3.99	1.25	0	5
52	Conflict within village	Conflict occurred last month	Continuous (0-7). Summation of 7 binary variables that indicate whether interviewee reports that last month there was a conflict of one of the following seven types in the village: Land conflict; Dowry conflict; Ethnic conflict; Act of burglary; Fights; Domestic violence; Murders or armed robbery	3702	0.99	1.29	0	7
53	Conflict within village	Number of conflicts occurred last month	Continuous. Summation of frequency of conflicts occurring last month, across the following seven types in the village: Land conflict; Dowry conflict; Ethnic conflict; Act of burglary; Fights; Domestic violence; Murders or armed robbery	3702	2.25	3.99	0	51
54	Conflict within village	Number of conflicts peacefully resolved	Continuous. Summation of frequency of peacefully resolved conflicts occurring last month, across the following seven types in the village: Land conflict; Dowry conflict; Ethnic conflict; Act of burglary; Fights; Domestic violence; Murders or armed robbery	3702	1.44	3.09	0	51
55	Conflict within village	Chief solved conflict	Continuous (0-7). Who solved the conflict: (1) The village chief; (2) The governor of the province; (3) The head of the territory (or his council); (4) The chief of the chieftainship (or his council); (5) The leader of the group (or his council); (6) The chief of the locality (or his council); (7) All the inhabitants of the village together; (8) Elders (committee of elders, community BARZA, etc.); (9) A local notable other than the village chief (teacher, doctor, etc.); (10) A religious authority; (11) An NGO; (12) The locally elected deputy for the provincial assembly; (13) The locally elected deputy for the national assembly in Kinshasa; (14) The government of Kinshasa (presidency of the Republic, a ministry, the Social Fund, etc.); (15) Police; (16) The army (FARDC); (17) The Monusco; (18) An armed group (mai-mai, raya, FDLR, ADF, LRA, etc.) Option (1) equals 1. Sum across seven types of conflict.	1825	0.44	0.74	0	4
56	Conflict within village	Other local actor solved the conflict	Continuous. Who solved the conflict: (1) The village chief; (2) The governor of the province; (3) The	1825	0.30	0.63	0	4

			head of the territory (or his council); (4) The chief of the chieftainship (or his council); (5) The leader of the group (or his council); (6) The chief of the locality (or his council); (7) All the inhabitants of the village together; (8) Elders (committee of elders, community BARZA, etc.); (9) A local notable other than the village chief (teacher, doctor, etc.); (10) A religious authority; (11) An NGO; (12) The locally elected deputy for the provincial assembly; (13) The locally elected deputy for the national assembly in Kinshasa; (14) The government of Kinshasa (presidency of the Republic, a ministry, the Social Fund, etc.); (15) Police; (16) The army (FARDC); (17) The Monusco; (18) An armed group (mai-mai, raya, FDLR, ADF, LRA, etc.) Options (4)-(10) equal 1. Sum across seven types of conflict.					
57	Conflict between village	Conflict occurred last month	Continuous (0-7). Summation of 7 binary variables that indicate whether interviewee reports that last month there was a conflict of one of the following seven types between the village and another village: Land conflict; Dowry conflict; Ethnic conflict; Act of burglary; Fights; Domestic violence; Murders or armed robbery	3702	0.50	0.98	0	6
58	Conflict between village	Number of conflicts occurred last month	Continuous. Summation of frequency of conflicts occurring last month, across the following seven types between the village and another village: Land conflict; Dowry conflict; Ethnic conflict; Act of burglary; Fights; Domestic violence; Murders or armed robbery	3702	1.06	3.18	0	67
59	Conflict between village	Number of conflicts peacefully resolved	Continuous. Summation of frequency of peacefully resolved conflicts occurring last month, across the following seven types between the village and another village: Land conflict; Dowry conflict; Ethnic conflict; Act of burglary; Fights; Domestic violence; Murders or armed robbery	3702	0.83	2.72	0	66
60	Conflict between village	Chief solved conflict	Continuous (0-7). Who solved the conflict: (1) The village chief; (2) The governor of the province; (3) The head of the territory (or his council); (4) The chief of the chieftainship (or his council); (5) The leader of the group (or his council); (6) The chief of the locality (or his council); (7) All the inhabitants of the village together; (8) Elders (committee of	995	0.19	0.46	0	3

			elders, community BARZA, etc.); (9) A local notable other than the village chief (teacher, doctor, etc.); (10) A religious authority; (11) An NGO; (12) The locally elected deputy for the provincial assembly; (13) The locally elected deputy for the national assembly in Kinshasa; (14) The government of Kinshasa (presidency of the Republic, a ministry, the Social Fund, etc.); (15) Police; (16) The army (FARDC); (17) The Monusco; (18) An armed group (mai-mai, raya, FDLR, ADF, LRA, etc.) Option (1) equals 1. Sum across seven types of conflict.					
61	Conflict between village	Other local actor solved the conflict	Continuous. Who solved the conflict: (1) The village chief; (2) The governor of the province; (3) The head of the territory (or his council); (4) The chief of the chieftainship (or his council); (5) The leader of the group (or his council); (6) The chief of the locality (or his council); (7) All the inhabitants of the village together; (8) Elders (committee of elders, community BARZA, etc.); (9) A local notable other than the village chief (teacher, doctor, etc.); (10) A religious authority; (11) An NGO; (12) The locally elected deputy for the provincial assembly; (13) The locally elected deputy for the national assembly in Kinshasa; (14) The government of Kinshasa (presidency of the Republic, a ministry, the Social Fund, etc.); (15) Police; (16) The army (FARDC); (17) The Monusco; (18) An armed group (mai-mai, raya, FDLR, ADF, LRA, etc.) Options (4)-(10) equal 1. Sum across seven types of conflict.	995	0.20	0.48	0	3
62	Inclusion	Attitudes towards internally displaced	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that internally displaced Live in village; Acquire land title; Become settlement chief; Become close friend; Marry a family member.	3368	4.16	1.04	0	5
63	Inclusion	Attitudes towards refugees	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that refugees Live in village; Acquire land title; Become settlement chief; Become close friend; Marry a family member.	3361	3.05	1.66	0	5
64	Inclusion	Attitudes towards ex-combatants	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that ex-combatants Live in village; Acquire land title; Become settlement chief; Become close friend; Marry a family member.	3356	3.25	1.80	0	5

65	Inclusion	Attitudes towards neighbouring villagers	Continuous (0-5). Summation of 5 binary variables that indicate whether interviewee accepts that neighbouring villagers Live in village; Acquire land title; Become settlement chief; Become close friend; Marry a family member.	3366	4.18	0.99	0	5
66	Economic Welfare	HH assets ownership	Continuous. How many of the listed good does a household own.: Cattle (sheep, goats, cows or pigs); Poultry (chickens, ducks, turkeys); Buildings Cases (house); Rooms (total); Beds / mattresses (foam, straw, sheets, linen or mat); Buckets; Basins; Kerosene lamp (or equivalent); Motorbike; Bicycles; Tsukudu; Car; Hoes and machetes; Saucepans; Wardrobe; Canoe; Camera; Radio; Television; DVD player; Cell phone.	3372	0.03	1.79	-3	13
67	Economic Welfare	HH consumption expenditures	Continuous. Amount spent in the last two weeks on the following goods and services: Food; Medical expenses; Hobbies; Clothing; Ornaments; Alcohol; Cigarettes; Seeds / fertilizers; Household furniture / equipment; Minor house maintenance; Major house maintenance. sum	3372	36260	288702	0	16000000
68	Income Generating Activities	Employment	Binary. Do you currently have an income-generating activity? This includes any form of income-generating activity, such as salaried employment, microenterprise, agriculture, day laborer, etc.	3362	0.73	0.44	0	1
69	Income Generating Activities	Working Hours	Continuous. During the past week (7 days) how many hours did you work on this income generating activity	2382	32.60	18.37	0	96
70	Income Generating Activities	Earnings of interviewee	Continuous. In the past two weeks how much money did you earn?	3151	53031	200382	0	7240000
71	Income Generating Activities	Earnings of household	Continuous. In the past two weeks how much money did your household earn?	3032	72818	212526	0	7240000
72	Subjective Well-Being	Self-Perception of life conditions	Continuous. Compared to other households in your village, how do you consider the economic conditions of your household? (1) Much worse, (2) Worse, (3) Better, (4) Much better	3355	2.56	1.23	1	5
73	Subjective Well-Being	School Attendance	Continous. Average per household. (Q43) In the last 2 weeks (15 days), how many days the household kids in your households went to school? For all the children.	2632	7.73	4.84	0	15
74	Subjective Well-Being	Illness	Binary. During the past 2 weeks have you or a member of your household become so ill that you cannot work (or go to school)? Reverse for the index	3372	0.55	0.50	0	1

TABLE A 5. VARIABLE DEFINITIONS AND SUMMARY INFORMATION FOR VARIABLES USED IN MANIPULATION CHECK

#	Family	Variable	Description	Obs.	Mean	Sd.	Min.	Max.
1	Manipulation Check	Knowledge of STEP	Binary (0-1). Do you know about the STEP project? (0) Do not know at all ; (1) Know remotely (heard of it but not who did it) (2) Know (the project but not the name STEP ; (3) Know it all Options 1, 2 and 3 equal 1.	3372	0.54	0.50	0	1
2	Manipulation Check	STEP was built	Binary (0-1). Was the project built?	3372	0.18	0.39	0	1
3	Manipulation Check	Satisfied with STEP	Continuous (0-18). Summation across 18 different categories on satisfaction. *4 positively formulated:* Are you satisfied with the initiative taken to build this structure? With the quality of the work? With the Building? With the collaboration between communities that it entailed? 0) totally disagree, (1) disagree, (2) agree, (3) totally agree. Options 2 and 3 equal 1. *14 negatively formulated (inversed before summation):* The project created divisions or conflicts between the communities The project created divisions or conflicts in the community The project did not reflect your concerns The selected projects did not help enough people in this village The selected projects were not the most important for this village There has been embezzlement in this village (corruption, nepotism, etc.) Disagreements about the project in this village were not well managed The allocation of money across villages was not done fairly The ALE (NGO in charge of the project) did not behave well in the village There was not enough information about the process The whole process (project decision and execution) took too long	1839	8.50	6.10	0	18

			The process required too many steps / formalities to follow I had no real influence on the selection of projects The project was controlled by the chief					
4	Manipulation Check	STEP use by respondent	Last week, how many times did you personally use the STEP infrastructure?	1368	1.64	4.02	0	30
5	Manipulation Check	STEP use by household	Last week, how many times did you personally or your household members use the STEP infrastructure	1391	3.15	5.70	0	50
6	Manipulation Check	Project Identification	Categorical (0-4). Participated in identifying needs of community. Summation over four possible aid projects (most respondents mention 0 or 1 project).	3372	0.14	0.40	0	4
7	Manipulation Check	Project Contribution	Categorical (0-4). Participated in project contribution mobilization. Summation over four possible aid projects (most respondents mention 0 or 1 project).	3372	0.14	0.38	0	3
8	Manipulation Check	Project Execution	Categorical (0-4). Participated in project execution. Summation over four possible aid projects (most respondents mention 0 or 1 project).	3372	0.19	0.45	0	3
9	Manipulation Check	Project Administration	Categorical (0-4). Participated in project administration. Summation over four possible aid projects (most respondents mention 0 or 1 project)	3372	0.08	0.32	0	3

Appendix 3: Deviations from the Pre-Analysis Plan

This study was pre-registered in the AER registry (DOI: 10.1257/rct.7602-1.0). In what follows, we briefly summarize deviations from what we set out to do. Table A 6 lists deviations from the pre-specified outcome variables

TABLE A 6. CHANGES IN OUTCOME MEASURES

Family	Outcome	Description	Action
Infrastructure access	Household needs infrastructures	Continuous (0-5). Which of the following types of infrastructure does your household need (if currently absent in locality)? (1) well, (2) bridge, (3) pipes, (4) (meeting) hall, (5) church.	Dropped because there is almost no variation: almost everybody says they need the infrastructure when it is lacking
Health infrastructure	Household needs health centre	Binary. Does your household need a health centre (if currently absent in locality)?	Dropped because there is almost no variation: almost everybody says they need a health centre when it is lacking
Health infrastructure	Fell ill during past 2 weeks.	Binary. During the past 2 weeks have you or a member of your household become so ill that you cannot work (or go to school)?	Moved from the outcome family 'health infrastructure' to well-being.
Education infrastructure	Household needs primary school	Binary. Does your household need a primary school (if currently absent in locality)?	Dropped because there is almost no variation: almost everybody says they need a primary school when it is lacking
Education infrastructure	Household needs secondary school	Binary. Does your household need a secondary school (if currently absent in locality)?	Dropped because there is almost no variation: almost everybody says they need a secondary when it is lacking
STEP	Knowledge of STEP STEP was built Satisfied with STEP STEP use by respondent STEP use by household Project Identification Project Contribution Project Execution Project Adiminstration		These variables were kept but used in a manipulation check rather than as part of an outcome index.
Cleavages	Ranking of top-3 cleavages within village that drive violent conflicts	Ranking of three most important cleavages in society, to be chosen from cleavages mentioned as occurring in village (in a preceding question in the questionnaire): between the richest and the poorest; men and women; young	Dropped because these variables could not be coded in a meaningful way, as they were conditional on villages first indicating that there was a problem related to the specific

		and old; natives and newcomers; between religions; tribes or ethnic groups; and between different elites).	division. Hence, these variables were only informative of the within-village ranking of conflicts, not so much the between-village ranking of conflicts. We replaced them by variables in next row
Cleavages	Cleavages driving violent conflicts	Continuous (0-7) (Out of a list of seven different divisions:) Did these divisions drive violent events in the last 6 months. Binary. 0 No 1 Yes. Seven divisions concerned: between the richest and the poorest; men and women; young and old; natives and newcomers; between religions; tribes or ethnic groups; and between different elites Sum across seven types, and replace missing by 0 as this indicates that this type of conflict did not take place (and could therefore not turn violent).	Added, in replacement of the variables in the row above

Appendix 4: Regression results for each individual outcome variable

TABLE A 7. REGRESSION RESULTS FOR EACH INDIVIDUAL OUTCOME VARIABLE

Family	Variable	Control	CDD (Se)	CDD + (se)	N
Infrastructure access	INDEX	0.00	-0.17 (0.13)	-0.09 (0.13)	3362
	Presence of well	0.37	0.00 (0.06)	-0.03 (0.06)	3353
	Presence of bridge	0.37	-0.12** (0.06)	-0.09 (0.06)	3349
	Presence of pipes	0.38	0.03 (0.07)	0.10 (0.06)	3348
	Presence of (meeting) hall	0.17	-0.08** (0.04)	-0.08** (0.04)	3337
	Presence of church	0.95	-0.02 (0.03)	0.01 (0.03)	3360
	Infrastructures with high-quality wall	0.70	-0.05 (0.08)	-0.09 (0.07)	3362
	Infrastructures with high-quality roof	0.99	-0.08 (0.07)	-0.05 (0.06)	3362
Infrastructure use	INDEX	0.00	-0.24 (0.18)	-0.44** (0.18)	1174
	Frequency of use of infrastructures	36.7	-1.07 (3.81)	-0.58 (3.97)	3201
Health infrastructure	INDEX	0.00	-0.04 (0.14)	-0.18 (0.14)	3360
	Presence of health centre	0.81	-0.02 (0.05)	-0.07 (0.05)	3360
	Health center with high-quality wall	0.62	0.01 (0.07)	0.06 (0.06)	2476
	Health center with high-quality roof	0.90	0.02 (0.04)	0.04 (0.04)	2478
	Frequency of use of health infrastructure	2.66	-0.46 (0.39)	-0.39 (0.37)	2376
	Walking time to nearest health post	2.78	0.12 (0.12)	0.21* (0.12)	3313
	Walking time to nearest health centre	2.95	0.17 (0.12)	0.13 (0.11)	3355
	Walking time to nearest hospital	4.03	0.14 (0.19)	0.09 (0.17)	3333

	Seeks health care in health care facility	0.77	-0.05 (0.04)	0.02 (0.04)	3329
Education infrastructure	INDEX	0.00	0.07 (0.14)	0.08 (0.14)	3360
	Presence of primary school	0.92	0.02 (0.04)	0.02 (0.04)	3360
	Presence of secondary school	0.90	-0.06 (0.05)	-0.08 (0.05)	3359
	Primary school with high-quality wall	0.57	0.01 (0.06)	0.05 (0.06)	2939
	Primary school with high-quality roof	0.87	0.03 (0.04)	0.06 (0.05)	2951
	Secondary school with high-quality wall	0.57	-0.04 (0.07)	-0.04 (0.07)	2712
	Secondary school with high-quality roof	0.88	-0.06 (0.05)	-0.02 (0.05)	2724
	Walking time to nearest primary school	2.54	0.00 (0.08)	0.11 (0.09)	3356
	Walking time to nearest secondary school	2.67	0.13 (0.10)	0.16 (0.10)	3356
	School-aged children go to (high-quality) school	1.59	-0.13* (0.07)	-0.02 (0.06)	1990
Satisfaction STEP	INDEX	0.00		0.04 (0.13)	1350
	Satisfaction with the CDD process	9.96		0.19 (0.67)	1350
	Personal use of STEP project	2.89		0.70 (0.69)	1264
	Household use of STEP project	1.70		0.35 (0.46)	1239
Infrastructure provision – chief survey	INDEX	0.00	-0.09 (0.16)	-0.16 (0.14)	299
	Number of wells	3.37	-0.42 (0.72)	-0.75 (0.65)	299
	Number of (meeting) halls	0.32	-0.13 (0.11)	-0.08 (0.10)	292
	Number of churches	4.40	0.06 (0.55)	-0.43 (0.46)	297
	Number of health centers	1.54	-0.20 (0.20)	-0.31 (0.18)	299
	Number of schools (classes)	26.9	6.09 (4.16)	-0.55 (3.74)	290
	Number of infrastructure obtained on request of village	21.2	1.75 (3.76)	-4.38 (3.19)	303
	Number of infrastructure obtained at initiative of state of NGO without village contribution	10.6	-2.18 (2.54)	-0.99 (2.48)	303
	Number of infrastructure obtained at initiative of state of NGO with village contribution	4.06	2.38 (1.32)	2.50** (1.22)	303
Community organization	INDEX	0.00	-0.22* (0.13)	-0.14 (0.11)	2336
	Committee attended by HH	2.11	-0.21 (0.20)	-0.35* (0.21)	3315
	Frequent committee attendance by HH	18.8	-4.83* (2.83)	-3.17 (2.44)	2336
	Number of committees organized in village	1.70	-0.41*** (0.14)	-0.29* (0.15)	2581
	Frequency of committee meetings organized in village	1.70	-0.33** (0.13)	-0.24* (0.14)	2026
Information transmission	INDEX	0.00	-0.15 (0.12)	-0.05 (0.13)	3168
	Deviation between chief's and HH answers wrt aid received	0.12	-0.05 (0.04)	-0.02 (0.04)	3168
	Deviation between chief's and HH answers wrt source of aid received	0.03	0.03 (0.03)	0.07** (0.03)	3362
	Deviation between chief's and HH answers wrt amount of the aid received	1.52	-0.19 (0.65)	-0.06 (0.65)	3362
	Deviation between chief's and HH answers wrt perception of favoritism to other villages	0.93	-0.06 (0.27)	0.01 (0.26)	1501
Trust	INDEX	0.00	0.00 (0.11)	0.14 (0.11)	3361
	Intra-group trust	2.00	0.00 (0.09)	0.11 (0.09)	3361
	Inter-group trust	0.70	0.05 (0.11)	0.10 (0.11)	3361
Cleavages	INDEX	0.00	0.13 (0.12)	0.27** (0.12)	3362
	Within-village cleavages	0.75	0.13 (0.12)	0.27** (0.12)	3362
	Cleavages driving violent conflicts	0.56	0.10 (0.11)	0.21* (0.11)	3362
Cohesion	INDEX	0.00	-0.13 (0.11)	0.11 (0.09)	3358
	Attitudes towards returned internally displaced	4.42	-0.11 (0.08)	0.08 (0.08)	3358
	Attitudes towards returned refugees	4.11	-0.10 (0.13)	-0.04 (0.13)	3354
	Attitudes towards individuals of other ethnic groups	4.03	-0.11 (0.13)	0.10 (0.10)	3352
	INDEX	0.00	-0.12 (0.11)	-0.14 (0.12)	3665

Conflict within village	Conflict occurred last month	1.05	-0.16 (0.15)	-0.19 (0.16)	3665
	Number of conflicts occurred last month	2.53	-0.8* (0.44)	-0.62 (0.44)	3665
	Number of conflicts peacefully resolved	1.51	-0.27 (0.31)	-0.28 (0.32)	3665
	Chief solved conflict	0.47	0.06 (0.11)	0.17 (0.12)	1808
	Other local actor solved the conflict	0.40	-0.16* (0.09)	-0.19** (0.09)	1808
Conflict between village	INDEX	0.00	0.05 (0.13)	-0.01 (0.14)	3665
	Conflict occurred last month	0.51	0.05 (0.12)	-0.01 (0.13)	3665
	Number of conflicts occurred last month	1.03	0.14 (0.34)	0.26 (0.32)	3665
	Number of conflicts peacefully resolved	0.79	0.25 (0.31)	0.21 (0.27)	3665
	Chief solved conflict	-0.21	0.07 (0.06)	-0.05 (0.09)	984
	Other local actor solved the conflict	-0.24	0.12 (0.08)	0.04 (0.08)	984
Inclusion	INDEX	0.00	-0.16 (0.11)	-0.02 (0.11)	3359
	Attitudes towards internally displaced	4.20	-0.14 (0.10)	-0.02 (0.09)	3359
	Attitudes towards refugees	3.21	-0.20 (0.17)	-0.17 (0.18)	3352
	Attitudes towards ex-combatants	3.21	-0.18 (0.19)	-0.04 (0.19)	3347
	Attitudes towards neighbouring villagers	4. 20	-0.20* (0.11)	0.11 (0.08)	3357
Economic Welfare	INDEX	0.00	0.11 (0.09)	0.01 (0.08)	3362
	HH assets ownership	0.06	0.19 (0.16)	0.02 (0.13)	3362
	HH consumption expenditures	6.96	0.24 (0.38)	0.19 (0.37)	3362
Income Generating Activities	INDEX	0.00	0.05 (0.09)	0.00 (0.09)	3353
	Employment	0.75	0.02 (0.04)	0.00 (0.04)	3353
	Working Hours	35.2	-1.22 (1.75)	0.86 (1.77)	2379
	Earnings of interviewee	9.02	0.17 (0.33)	0.64** (0.30)	3142
	Earnings of household	10.1	-0.07 (0.27)	0.44* (0.26)	3025
Subjective Well-Being	INDEX	0.00	-0.09 (0.08)	-0.06 (0.08)	3362
	Self-Perception of life conditions	2.69	-0.06 (0.11)	-0.03 (0.12)	3346
	School Attendance	8.27	-0.98** (0.50)	-1.00** (0.48)	2627
	Illness	0.60	-0.04 (0.04)	-0.03 (0.04)	3362